

Taylor & Francis
Taylor & Francis Group

Applied Artificial Intelligence
An International Journal

ISSN: 0883-9514 (Print) 1087-6545 (Online) Journal homepage: www.tandfonline.com/journals/uaai20

Towards Explainable Machine Learning for Prediction of Disease Progression

Stijn Berendse, Johannes Krabbe, Jonas Klaus & Faizan Ahmed

To cite this article: Stijn Berendse, Johannes Krabbe, Jonas Klaus & Faizan Ahmed (2024) Towards Explainable Machine Learning for Prediction of Disease Progression, Applied Artificial Intelligence, 38:1, 2423510, DOI: 10.1080/08839514.2024.2423510

To link to this article: https://doi.org/10.1080/08839514.2024.2423510

© 2024 The Author(s). Published with license by Taylor & Francis Group, LLC.

View supplementary material

Published online: 07 Nov 2024.

Submit your article to this journal

Article views: 1774

View related articles

View Crossmark data

Citing articles: 1 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=uaai20

APPLIED ARTIFICIAL INTELLIGENCE
2024, VOL. 38, NO. 1, e2423510 (44 pages)
https://doi.org/10.1080/08839514.2024.2423510

Taylor & Francis
Taylor & Francis Group

OPEN ACCESS Check for updates

Towards Explainable Machine Learning for Prediction of Disease Progression

Stijn Berendse[a], Johannes Krabbe[b], Jonas Klaus[c], and Faizan Ahmed[a]

[a]Department of Computer Science, University of Twente, Enschede, The Netherlands; [b]Unilabs/Medisch Spectrum Twente Enschede, The Netherlands; [c]Asklepios Klinik Barmbek Hamburg, Germany

ABSTRACT

This research focuses on addressing the challenges surrounding interpretability of machine learning techniques in the field of prediction of disease progression. This paper summarizes the state-of-the-art in machine learning for disease progression modeling and challenges related to this context. Based on this state-of-the-art, we design and develop a pipeline consisting of data preparation, prediction, and explanation. Predictions are made using a deep recurrent neural network-based model which is followed by an integration of the LIME framework to provide explanations for each prediction. We demonstrate our pipeline by applying it to two diverse case studies on diabetes and Parkinson's disease. Besides this, we compare the influence of three data imputation methods on predictive performance. Results of the comparison show that there is no statistically significant difference in performance due to different data imputation techniques. Furthermore, we provide a number of concrete recommendations and directions for future research, such as improving input flexibility of the prediction model and improving the visualization of generated explanations. Based on the results of this research, we conclude that the proposed pipeline achieves the goal of integrating a state-of-the-art prediction model and the LIME framework to make prediction of disease progression more transparent and interpretable.

ARTICLE HISTORY
Received 17 June 2024
Revised 12 September 2024
Accepted 19 October 2024

Introduction

As machine learning and artificial intelligence (AI) have become widespread in their use Jordan and Mitchell (2015) and are reaching ever more sensitive domains such as healthcare, the need for responsible AI has arisen Vayena, Blasimme, and Glenn Cohen (2018); Hunter and Holmes (2023). AI systems work like a "black box:" they receive input data, train a model, and then use that model to make predictions.

However, this nontransparent approach has major drawbacks, including limited or missing insight into the reasoning behind the prediction which

CONTACT Faizan Ahmed, faizan.ahmed@utwente.nl Department of Computer Science, University of Twente Enschede, Hallenweg 19, Enschede 7522 NH, the Netherlands

© 2024 The Author(s). Published with license by Taylor & Francis Group, LLC.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

leads to a deficiency in trust and raises ethical concerns among clinicians Gunning and Aha (2019). Due to the need for responsible AI, previously waned interest in the field of XAI (eXplainable AI) re-surged. In addition to generating predictions based on input data, XAI is designed to provide a way of explaining how and why a certain prediction was made Tjoa and Guan (2021).

Disease progression modeling is a concept that has proven useful in clinical management of diseases, particularly chronic disease (Mould (2012)) such as Alzheimer's disease (Ito et al. (2010)) and diabetes (Post et al. (2005)). Modeling disease progression enables physicians to predict patient-specific disease trajectories, that can guide personalized therapeutic strategies. Research has shown that machine learning performs well in the field of disease modeling Papaiz et al. (2022). However, the interpretability of such models poses a challenge. To ensure that machine learning-based prediction of disease progression and subsequent decision-making is reliable and ethical, it is essential to improve the transparency and interpretability of the models used. Explainability can improve the interaction between clinicians and prediction systems by ensuring clinicians feel more confident in leveraging the power of prediction models – an interaction that is essential for the adoption and integration of such systems in healthcare.

In this paper, we first review the state-of-the-art in machine learning for disease progression modeling. Based on the review, we select a prediction model and extend it with a degree of explainability using the well-known LIME framework and a minor proposed extension of the LIME Python implementation. Consequently, we apply this explainable pipeline to two case studies. The first case study uses a dataset from the Parkinson's Progression Markers Initiative (PPMI). The second case study is performed with a dataset on diabetes patients collected in the Netherlands.

The primary contribution of this research is a deep recurrent neural network-based pipeline, capable of predicting disease progression based on time-series data, whilst providing explanations for its predictions through an integration of the LIME framework. Additionally, in the design phase of this pipeline, we perform an experiment to determine what missing data imputation technique leads to the best predictive performance. We demonstrate the pipeline's functionality by applying it to two case studies with real patient data: one case study with a small dataset, the other with a large dataset.

This research does not focus on developing new algorithms, but rather on integrating existing components to demonstrate how current techniques can be used together to allow clinicians and AI to collaborate more effectively. This contribution advances the field of machine learning for prediction of disease progression by showcasing the importance and potential of integrating interpretability into state-of-the-art prediction pipelines.

We conclude this section by providing the structure of this paper. In Section 2, we explore the state-of-the-art in prediction of disease progression and related topics. Subsequently, in Section 3 we describe the design and validation of our proposed pipeline using two case studies. Section 4 contains the results for the performed experiments per case study. Next, in Section 5, we discuss limitations of this research and provide directions for future research. Finally, concluding remarks are presented in Section 6.

Background

This section provides background on the state-of-the-art in predicting disease progression, as well as explainability frameworks. Furthermore, we discuss disease severity proxies for Parkinson's disease and diabetic kidney disease.

Prediction Models and Data Types

Research on prediction of disease progression can be broadly categorized into two classes: milestone prediction and trajectory prediction. Milestone prediction focuses on predicting the occurrence of a single event, such as conversion from a lower disease severity level to a higher severity level, or the death of a patient. Trajectory prediction is based on predicting biomarkers, disease severity or other disease-related features over time.

Within milestone prediction, there are two primary types of prediction outcomes: binary predictions (for example, whether a patient will or will not progress to a more severe disease state) and predictions of event probabilities. Binary predictions are often made using fairly simple techniques such as random forests and support vector machine (see e.g. Cervantes et al. 2020). Three studies compare these two techniques, with SVM outperforming random forests in both that report performance scores Yılmaz Acar, Başçiftçi, and Ekmekci (2022); Seccia et al. (2021); El-Sappagh et al. (2022). Prediction of event probability is often performed using a Cox Proportional Hazards model Cox (1972), typically combined with Kaplan-Meier curves for visualization Nazha et al. (2021); Chen et al. (2021); Schmidt-Erfurth et al. (2018); Taghavi et al. (2021); Eng et al. (2019).

The second identified prediction goal is trajectory prediction. Trajectory prediction aims to predict the exact condition of patients at a specific time point or over time. In trajectory prediction, simpler models such as SVM and random forests are far less popular than in milestone prediction. Instead, more complex techniques are used, such as two-step systems Dadu et al. (2022); Bhagwat et al. (2018); Sadiq et al. (2022); Schulam and Saria (2016). These systems largely consist of an initial step in which patient subgroups are created, in order to form trajectory clusters or phenotypes, followed by a second step that assigns patients to one of these subgroups.

While milestone predictions offer decent performance using simple (and thus interpretable) techniques, they may offer less value for supporting clinical decision-making than trajectory predictions. Milestone predictions are mostly binary, only stating whether a certain event will take place, but not when or to what extent. Trajectory predictions may offer this additional information, especially being more specific regarding the severity over time. One of the limitations of trajectory prediction is that a good proxy for disease severity must be found. For some diseases, this can be a measurable datapoint, such as the forced expiratory volume functioning as a proxy for pulmonary exacerbation in cystic fibrosis patients Su, Wang, and Szczesniak (2020). In other cases, such as Alzheimer disease, disease severity is assigned based on the results of cognitive testing Jiang et al. (2019). Despite this limitation, the enhanced quality and precision provided by trajectory prediction made it the preferred foundation for our pipeline.

A notable observation regarding input data type is that a majority of literature, regardless of whether the aim is milestone prediction or trajectory prediction, uses single data timepoints as model input. Studies that utilize time-series data tend to use more specialized models adapted to this task, particularly those with some form of internal recurrence. Examples of such models include Recurrent Neural Network (RNN) based models and its various subtypes. A popular RNN subtype is Long Short-Term Memory, or LSTM for short. This model type is used in a number of studies (Sun et al. (2021); Seccia et al. (2021); Ho et al. (2022); El-Sappagh et al. (2022); Jung et al. (2021); Larie, An, and Cockrell (2021)), but some studies propose combined models consisting of LSTM in conjunction with another model. Examples include LSTM-CNN An, Tang, and Wang (2021) and LSTM-autoencoder Chen et al. (2022). De Brouwer et al. (2021) and Dang et al. (2022) opt for a different RNN: Gated Recurrent Units, or GRU for short. GRU is characterized by a slightly less complex internal structure than LSTM, meaning it performs better on smaller datasets and with less computational power at the cost of performance degradation for large datasets.

A drawback of time-series data is that it requires more input data to train a model compared to single timepoint input. This is because each patient creates a single longitudinal training datapoint, whereas single timepoint-based systems can use each separate visit as a training datapoint. However, Bhagwat et al. (2018) compared these two approaches and concluded that models using time-series data offer better predictions than those using single timepoints as input. Therefore, to achieve optimal performance, we focus on developing a pipeline using time-series data as input.

Data Quality in Healthcare

Data in the healthcare domain, like other domains, suffer from various quality issues. The most commonly encountered problem is data sparsity, also known as missing data. This is because the measurements (e.g. biomarkers) taken are not necessarily consistent throughout various visits, even for the same patient. One of the most commonly used strategies for mitigating missing data is the use of some form of imputation El-Sappagh et al. (2022); Chen et al. (2021); Dadu et al. (2022); Sun et al. (2021); Ho et al. (2022); Jung et al. (2021); Marti-Juan, Sanroma-Guell, and Piella (2020); Sun et al. (2019); Chan et al. (2021); Dinov et al. (2016); Papaiz et al. (2022); Sharma et al. (2021); Zhang et al. (2021). Imputation techniques vary from longitudinal imputation Dadu et al. (2022); Jung et al. (2021); Sun et al. (2019) to cluster-based techniques (El-Sappagh et al. (2022)).

A well-known limitation of imputation is bias resulting from filled values that are non-representative of the truth (Chen et al. (2022)). Two methods have been proposed to mitigate this issue: signaling missing values in a separate vector Liu et al. (2020), or performing experiments both with and without imputation to assess an impact of imputation (Ma et al. (2021)). Schmidt-Erfurth et al. (2018) and Larie, An, and Cockrell (2021) avoid data sparsity issues by using a synthetic dataset and a dataset recorded for their research respectively. Other researchers used exclusion of features or data points as an approach to mitigate data sparsity (El-Sappagh et al. (2022); Dadu et al. (2022); Sun et al. (2019)). While this approach has no risk of bias from incorrect imputation, we note that exclusion-based mitigation does potentially remove relevant data from a dataset, which may reduce predictive performance.

A different challenge encountered in literature regarding prediction of disease progression is temporal irregularity. This refers to the inconsistency in the timing of sample collection from patients as visualized in Figure 1.

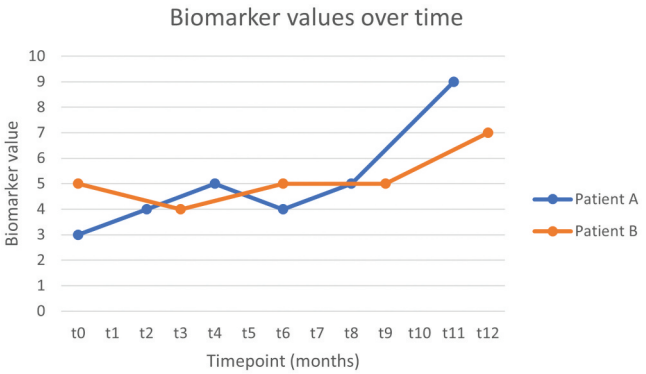


Figure 1. An example of temporal irregularity in biomarker sampling.

A problem which is exclusive to models that rely on time-series data rather than single time-points to predict disease progression.

Various handling methods are proposed, which can be categorized into two main categories. The first strategy involves handling by creating a robust model and the second is handling by pre-processing. This division is supported by Marti-Juan, Sanroma-Guell, and Piella (2020), who state that "temporal alignment" can be approached in two ways: pre-processing for a specific progression model and approaching this as a standalone problem. The first approach generally uses temporal characteristics to aid in integrating some form of a decay function An, Tang, and Wang (2021); Ho et al. (2022); Sun et al. (2021); De Brouwer et al. (2021); Jung et al. (2021) in a newly proposed model. Here, pre-processing is done for the specific model being developed. The latter often concerns some form of weighting time-points based on temporal characteristics Sun et al. (2019); Zheng and Hu (2020) or including temporal characteristics as a (normalized) feature to be interpreted by the model He, Huang, and Qian (2019); Sharma et al. (2021). These approaches are more model-agnostic, which fits the standalone problem concept.

Chen et al. (2022) adopted a unique method to address issues related to irregular timing of patient visits. They chose to ignore any extra visits that occurred between the first (or "baseline") visit and the last visit. This approach ensured that each patient had an equal number of visits for the training process. However, this method has a downside: it results in the loss of potentially valuable information that could be gained from including all visits. Similar to data sparsity issues, Schmidt-Erfurth et al. (2018) and Larie, An, and Cockrell (2021) managed to avoid the problems of irregular timing altogether by selecting datasets that lacked this issue.

LIME

Several frameworks are available to explain complex machine learning models including LIME. The LIME framework Ribeiro et al. (2016) is a method for explaining the predictions of any machine learning model, irrespective of its complexity or architecture. This model-agnostic nature means that LIME is a rare instance of an explainability framework that is applicable to time-series forecasting models.

LIME's core idea is to create a surrogate model that approximates the original model locally, using a simple, inherently explainable model such as a linear regression or a decision tree. Global interpretation can only be achieved by taking several local explanations, and analyzing them manually (Dieber and Kirrane (2020)). The surrogate model is trained on a set of perturbed instances that are generated by randomly tweaking some features of the original instance, and obtaining the predictions of the original model for these instances. These perturbed instances are then weighted by their

similarity to the original instance, so that instances that are more similar to the original have more influence on the surrogate model. The features and weights of the surrogate model are then used as the explanation for the prediction of the original model. As such, the explanation offered by LIME is merely an approximation of the true inner process of the model to be explained. However, due to the similarity check between perturbances and the original instance, this approximation holds for the local domain that LIME attempts to explain.

Ribeiro, the primary author of the LIME paper, offers a Python implementation of their framework on GitHub.[1] This LIME Python implementation includes support for time-series forecasting with RNNs, using the Recurrent Explainer classes. This makes the framework suitable for time-series forecasting interpretation. Each data sequence is split in both features and timepoints, so the influence of each feature at each timepoint is calculated and visualized separately. The python implementation does this by reshaping the input data to a format similar to $[val_1_{t1}, val_1_{t2}, val_1_{t3}, \ldots, val_2_{t1}, \ldots, val_n_{tm}]$. This format is interpretable by LIME due to the model-agnostic approach described above, so after the reshaping process, it can be fed into the regular interpreting code.

Disease Severity Proxies

Trajectory prediction models require a reliable proxy for disease severity. We perform case studies on two diseases, Parkinson's disease (PD) and diabetes, for which we must define this proxy.

To monitor the progression and condition of PD patients, various assessments have been developed. The most widely used assessment for PD is the Movement Disorders Society Unified Parkinson Disease Rating Scale (MDS-UPDRS) published in 2008 Goetz et al. (2008). The MDS-UPDRS consists of four partitions, assessing both motor and non-motor aspects of the disease through patient-reported information and observations by the clinician. Part one is focused on non-motor experiences in daily life, whereas the part two evaluates motor experiences in daily life. Part three is a motor examination, and part four is aimed at any motor complications. The cumulative score across all parts of the assessment serves as the proxy for overall disease severity in a patient in this study.

For the second case study, we focus on diabetic kidney disease, a condition associated with diabetes. Diabetic kidney disease (DKD) develops in 30% of patients with diabetes type 1 and 40% of patients with diabetes type 2, the overall prevalence of DKD is high Alicic, Rooney, and Tuttle (2017). Additionally, DKD is associated with an increased mortality Wen et al. (2017). There is a specific biomarker related to the development of diabetic kidney disease (DKD): the estimated Glomerular Filtration Rate (eGFR) of the

kidneys, measured in mL/min/1.73 m2. This biomarker is used to monitor the development of DKD in diabetes patients. An equation to classify chronic kidney failure based on the eGFR was developed in 1999 and refined in 2009, in the form of the Glomerular Filtration Rate estimation equation by the Modification of Diet in Renal Disease study group (GFR-MDRD) Levey et al. (1999) and Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) Levey et al. (2009) respectively. We use the eGFR value as the disease severity proxy for DKD.

Methodology

This section is split in three subsections. The first section discusses the general approach, model architecture, and a number of settings that are used to predict disease progression. Given two case studies, each further section describe the approach for that specific case study. Each case study section is divided in three parts. In the first part of each section, we will give a general overview of the dataset that is used and subsequently discuss the contents of the dataset. The second part will discuss the data preparation process, followed by a description of the model setup as the third and final part.

As mentioned in Section 1, before including the explanation phase in our pipeline, we first perform an experiment where we compare the influence of various imputation techniques on the performance of the predictive model. The compared imputation techniques are mean imputation, event mean imputation, and forward/backward filling. In this comparison, population mean imputation serves as the baseline due to its simplicity and lack of accommodation for any specific characteristics of patient data. The best performing imputation technique is used with the full pipeline, including explanation of predictions.

Pipeline Architecture

The architecture of our machine learning pipeline consists of three phases: pre-processing, prediction, and explanation generation. Figure 2 shows a high level overview of the pipeline while an in depth description is provided in the following subsections. The pre-processing is tied closely to the case studies thus the details of exact pre-processing steps are given together with the case studies.

The prediction model is the same for both case studies that consist of seven layers. The first two layers are GRU layers, as these recurrent layers serve to handle the longitudinal nature of the input data. After these two GRU layers, the shape of the data is still three-dimensional. Because this is a very high amount of "effective" features (sequence length $*$ number of features), we add an attention layer to prioritize important features and

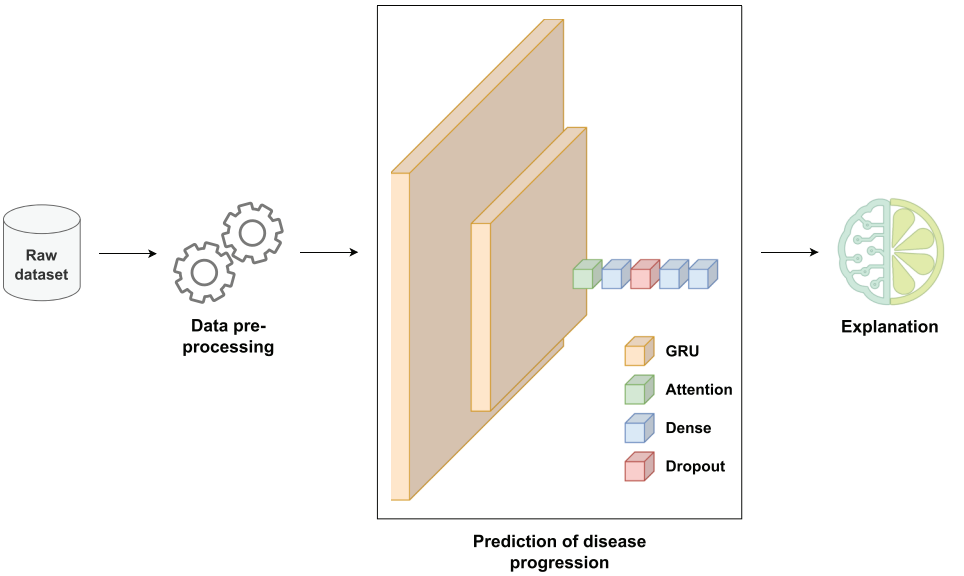


Figure 2. A high level overview of the proposed explainable pipeline.

timepoints. After the attention layer, we place the first dense layer, followed by a dropout layer to prevent overfitting. The final two layers are both dense layers.

As mentioned before, GRU and LSTSM are two variants of RNNs, which are neural network structure suitable for modeling sequential data. We choose GRU over LSTM because the data sequences in our case studies are relatively short. As discussed in Section 2, this suits GRU better than LSTM.

Pipeline Implementation

The implementation of the pipeline is done in Python. For the first part of the pipeline, data preparation, we use the Pandas library. For creating and training a predictive model, we use the Keras ML library, as it offers ready to use implementations of advanced techniques such as GRU layers. For the attention layer, we use a Keras-based implementation that is available on GitHub.[2] This implementation was chosen to ensure seamless integration between model layers. To convert the pre-processed data to a format that is compatible with the implementation of the predictive model, two actions are performed. The first is adding a column with ``target'' values by taking the value of the target variable at the next visit. The second is normalizing the training features. Finally, for the explainability phase of the pipeline, we use the Python implementation of the LIME framework as provided by Ribeiro, the primary author of the LIME paper (see Subsection 2.3 for details on this implementation). Because the Python LIME implementation is not capable of reverting normalized values to human-readable values, we provide an

extension script for the Python LIME implementation to perform this inversion before plotting the explanations.

A common issue we find in literature is that papers are published without code or details on implementation, thus hindering reproducibility. For the sake of transparency, we provide access to the code used to perform this research. Because the data we use is not freely available, it is not possible to share this along with the code. All notebooks and scripts used or written for this research can be found on Github.[3]

The optimizer used during the training of the predictive model is Keras' Adam optimizer, based on the paper by Kingma and Ba (2015). We choose Adam because of its relatively low computation time and low amount of required hyperparameter tuning. We opt for discretization of continuous values in LIME, because this leads to better explanations according to Dieber and Kirrane (Dieber and Kirrane (2020)). Furthermore, we choose to include the top 20 features in the explanation outputs.

Model Evaluation

The metric used to evaluate the model is the mean squared error, or MSE. The reason why we choose for the MSE rather than mean absolute error (MAE) is that MSE is harsher on larger errors. In the healthcare domain, the greater the inaccuracy of a prediction is, the greater the (adverse) effects can be. As such, we decided to use a metric that would penalize such mistakes harder than smaller mistakes. Furthermore, we use an 85%–15% train-test split, as well as K-fold cross validation in the training phase. Hence, for each experiment, five loss values will be reported: one per trained model. Along with these five loss values per experiment, we report the average loss across the five models as well as the standard deviation. Finally, since the splitting of the data and fold creation is random, seeds are used to ensure the research is reproducible. To account for possible imbalance in the data beyond the K-fold cross validation, we also perform our comparative experiment with the predictive model on four different seeds. These seeds are 787, 1998, 25 and 959,143, in no particular order.

Case Study 1: PPMI

The first case study is performed using the PPMI dataset. The dataset we use in this research includes visits up to July 7 2023. The PPMI is a structured study project, as a study of this size requires extensive planning of when the participants visit and what assessments and measurements are taken during the appointments. This structured nature is beneficial for this research, as visit intervals are fairly constant, and visit record contents are pre-determined. Therefore, we do not account for time irregularity in this dataset. Visits may be a few weeks off from perfectly yearly, but we explicitly assume that this

irregularity is not significant for the time scale at which the PPMI operates. The assessments that we choose to include are all parts of the MDS-UPDRS, as well as the Epworth Sleepiness Scale.

At the time of downloading the PPMI dataset for this research, the raw dataset contains 3199 participants, of whom 1558 (48.7%) are diagnosed with Parkinson's disease. In this population, there is basic demographic information for 3076 participants, of whom 1474 (47.9%) are PD patients. Out of the 3076 participants we have information on, 57.2% is male and 42.7% is female. For 0.16% of the participants, the gender is unknown. Figures 3 and 4 show the distribution of age for the full participant population and PD patient population in the raw dataset. Between these plots, we see little difference for the male population. When looking at the female population, we see a more uniform spread of ages for the PD patients compared to the plot containing all female PPMI participants. We note that the change in distribution appears to be caused by a change in the 55 to 70 year old age group. This may be caused by a relatively high number of control participants in this age range. Because the target variable (UPDRS total score) is an aggregation of various columns that are created during the data preparation process, we do not have information on this before dataset preparation is completed.

Data Preparation

An overview of the data preparation phase of our pipeline is shown in Figure 5. As mentioned, the full script to perform preparation is provided on Github.



Figure 3. The age distribution for all participants in the raw PPMI dataset.

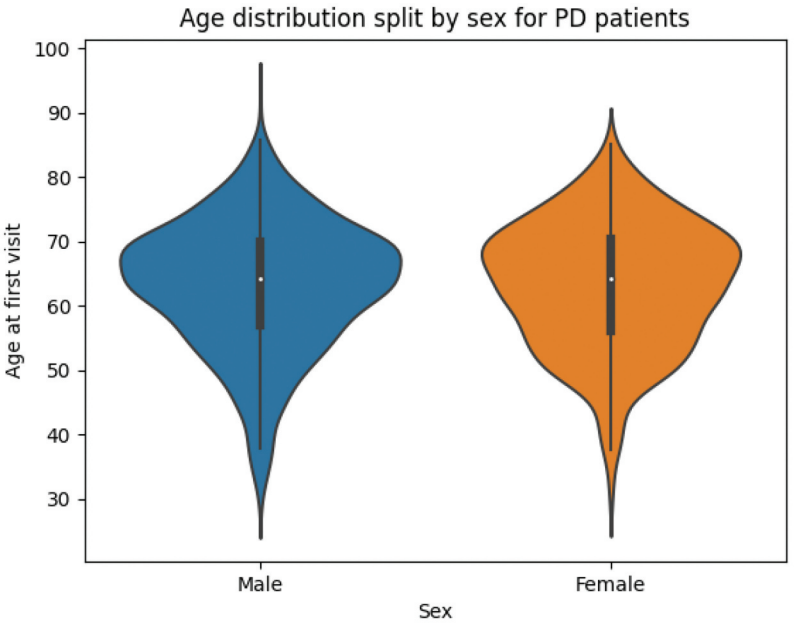


Figure 4. The age distribution for PD patients in the raw PPMI dataset.

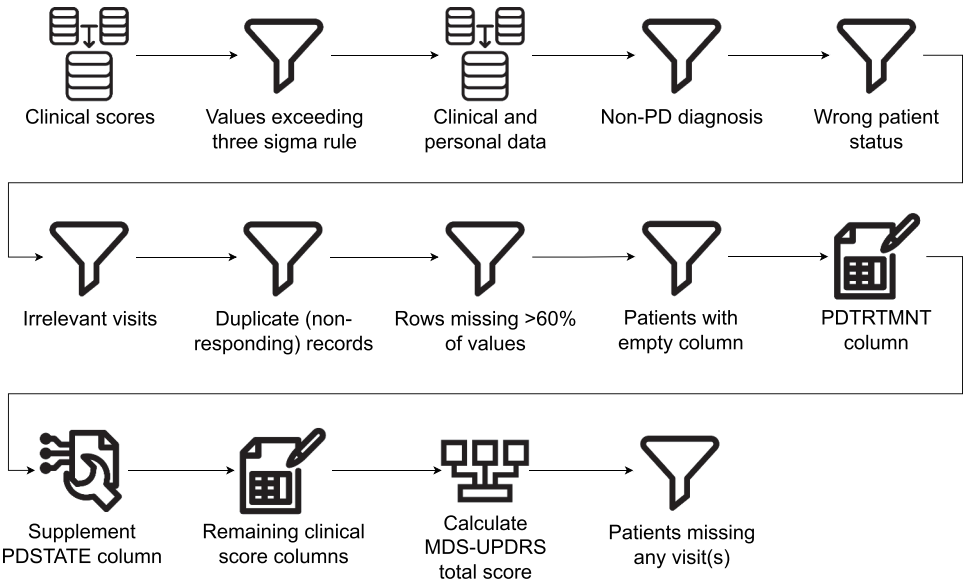


Figure 5. A flowchart of the pipeline's data pre-processing phase for the PPMI dataset. Icons taken from Freepik.

Due to the structured nature of the PPMI project, one can start by merging horizontally on a combination of patient ID and event ID for the six assessment CSVs. After merging, five columns containing the exact visit dates are removed. This results in a dataframe containing 22,749 rows and 82 columns,

covering 2774 participants. Once this is done, we remove any incorrectly registered values that fall outside the possible scale for the included assessments. Furthermore, any statistical outliers in the MDS-UPDRS partition total score columns are removed by applying the three sigma rule (see Pukelsheim 1994). This reduces the number of rows to 21,810 and the number of participants to 2725.

The next step is merging the participant data with the clinical scores, which is followed by the removal of any participants that do not have a PD diagnosis or that do not have a status "enrolled," "withdrawn," or "complete." The two columns containing this information are then discarded. Once more, the structured nature of PPMI is used, this time to select visits: only the first six yearly visits including the baseline visit are retained as these should register the same assessments for each patient. These steps significantly reduce the size of the data, but add two columns. At this point, 5812 rows and 84 columns remain, representing 1143 patients. Depending on the state of the patient during the assessment, which may be responding or non-responding to treatment, motor assessment scores especially may differ strongly. Because more patients have records where they are responding, and because some patients have two assessments linked to a single visit, one responding, one not responding, we choose to remove non-responding assessment records if a responding assessment record exists for the same visit. This leaves 4112 rows of data.

The next step is about handling missing data. It starts with some cleanup based on missingness. First, any records that are missing over 60% of values are removed as these records contain only a single, if any, complete MDS-UPDRS partition. Next, we remove any patients from the dataset that have a fully empty column across all visit records, because these patients cannot have this column imputed using forward/backward filling. Doing so removes another 880 rows, bringing the total down to 3232 rows and 716 patients. After this, we impute the column signifying whether a patient was undergoing treatment, and set their responsiveness/state according to this column to 0 or 1. If a patient is not in treatment during a visit, their state is set to 2 instead of the originally missing value. Once this is done, the remaining missing values are imputed using the chosen imputation method. Finally, the MDS-UPDRS partition total scores are summed in a new column to give the prediction target MDS-UPDRS total score and patients miss a visit record are removed to form the final pre-processed dataset.

Before preparation, the dataset contains 26,571 rows and 91 columns of data, covering 2774 participants. Once data preparation is completed, we are left with 1746 rows and 85 columns of data, covering 290 patients.

In Figure 6, the distribution of age and sex in the prepared dataset is shown. We see that in the male patient population, there is a fairly strong skewness toward higher age, whereas female patients have a fairly symmetrical age spread. Furthermore, we observe that the age range for the



Figure 6. The distribution of patient sex and age at the baseline visit in the prepared PPMI dataset.

female patients is greater than for male patients, although this appears to be caused by a very small number of young female patients. When we compare this distribution to the non-processed age distribution for PD patients shown in Figure 4, it appears that a relatively large number of patients just below the age of 70 is removed during the data preparation process, especially in the female population. The cause for this phenomenon could be that the raw dataset includes patients that have only had a screening for PPMI eligibility, but no baseline visit yet, whereas the prepared dataset does not. It would be plausible for a relatively large amount of screenings to be performed on patients around the age of 65 to 70, as this is the most common age at which PD is diagnosed Van Den Eeden et al. (2003). Furthermore, we see that the age range for male patients shrinks quite strongly, with the maximum age reducing from nearly 100 years to just over 85 years. This decrease appears to be caused primarily by an extremely high age outlier in the raw dataset that is removed in the preparation process.

A number of examples of patient MDS-UPDRS total score values over time are shown in Figure 7. We can see here that the progression is very heterogeneous: some patients do not seem to worsen over these first 5 years, whereas others display a very clear upward trajectory in disease severity, such as patients 193 and 270. Aside from this, Figure 7 clearly visualizes that progression may be highly fluctuating, as is the case with patients 40 and 274, for example.

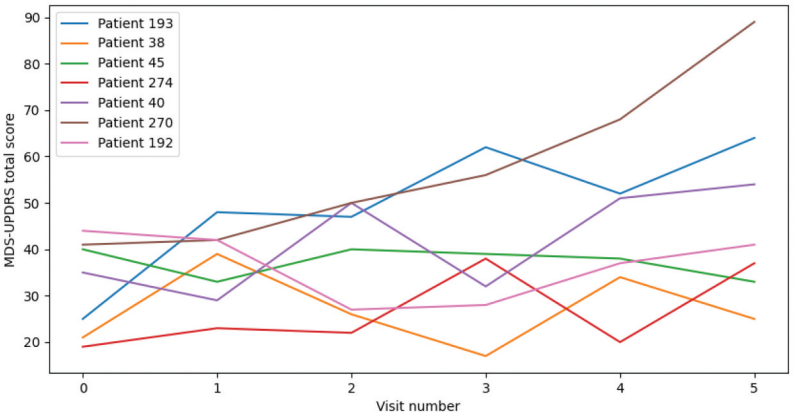


Figure 7. Several examples of patient MDS-UPDRS total score assessments over time.

Machine Learning Model

To train machine learning model, three hyperparameters must be set: the batch size, the learning rate and the dropout rate. Because the prepared dataset in this case study is small, we choose a relatively low batch size of 32, while leaving the model learning rate at the default 0.001. The dropout rate we choose for this case study is 0.2. As we have a small dataset in this case, it is likely that the model will overfit to the limited data it is trained on. By including the dropout at 0.2, we can reduce the degree to which the model (over)fits to the training data and increase the performance on unseen data.

Case Study 2: Diabetes

For the second case study, we use an anonymized dataset from presumed diabetes patients. This dataset is provided by Medlon, a medical laboratory based in Enschede, the Netherlands. To acquire this dataset, we signed a data transfer agreement. The time period covered by the dataset is from the start of 2015 to the first half of 2023. To gather the data for this dataset, records in which a HbA1c measurement was ordered in the laboratory request were selected from primary care providers such as general practitioners. Since HbA1c is used in routine monitoring of diabetes patients, assessed every three to six months, the time interval between the records is generally homogeneous. As such, we assume once again that temporal irregularity for this dataset is limited, and we do not account for such irregularity between data points. The dataset consists of the biomarkers that are included in these laboratory requests, such as the eGFR (a surrogate of kidney function) and blood glucose.

Due to the anonymization of this dataset, less demographic information is available than in our other case study. The sex of patients, for example, is not

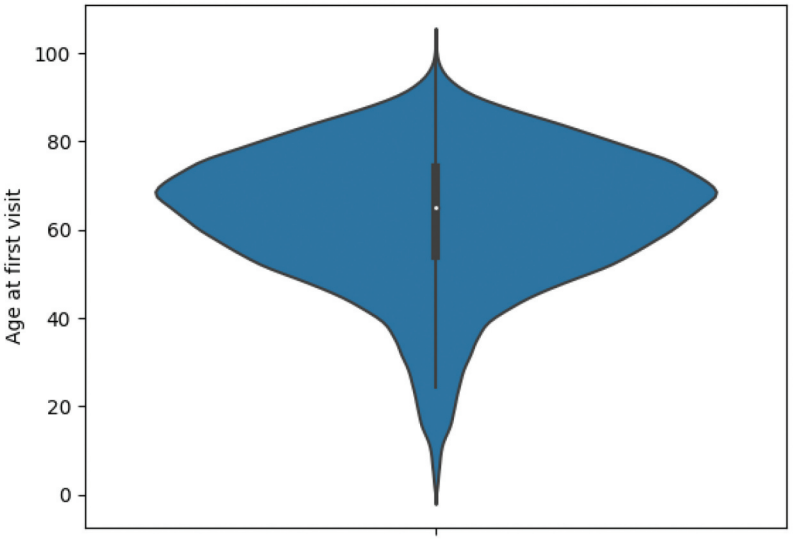


Figure 8. The distribution of patient age at the first visit in the raw dataset.

included in the data. Another piece of information that is not included in the dataset, is whether patients underwent medical interventions between visits that may strongly influence the eGFR either positively or negatively. Hence, it may be the case that the dataset contains large differences between visits that cannot directly be linked to patient's biomarker data.

The raw dataset on diabetes patients contains 128,793 patients. Figure 8 shows the distribution of age for the full population in the dataset. We see a strong peak in the number of patients around the age of 70. The prevalence of high age patients suggests that our dataset contains predominantly diabetes type 2 patients. Type 2 diabetes is usually developed at a later age, and around

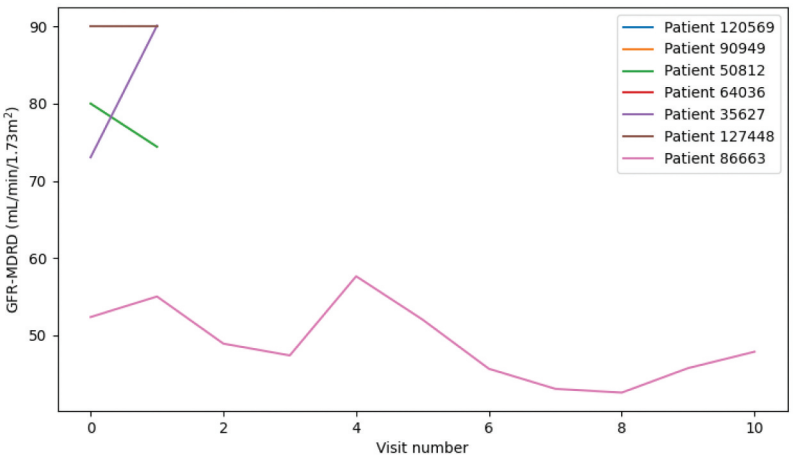


Figure 9. Several examples of patient eGFR assessments over time in the raw dataset.

95% of diabetes patients suffer from type 2 Association (2013), which supports our hypothesis on the distribution of diabetes types and the distribution of age in the dataset. In Figure 9, several random patient sequences are shown. It is clearly visible that the raw dataset is extremely sparse, with nearly half of the patients having no eGFR records at all, and only a single patient having more than two records. For this patient, with identifier 86,663, we see an irregular trend in eGFR, with one larger jump between visits 3 and 4.

Data Preparation

Similar to the previous case study, a high level overview of the data preparation pipeline is shown in Figure 10. Again, the full script is provided on Github.

As the diabetes dataset is split across files containing a year or half a year of data the first step is stacking the CSVs so they form one chronologically ordered dataframe. It resulted in a dataframe consisting of 737,010 rows and 861 columns. A total of 128,793 patients are included in this dataframe. Next, we remove any columns that are not relevant for progression of diabetes or specifically the prediction of DKD. Examples of such columns include whether the patient is allergic to a specific fruit, and the patient's desipramine blood content. The remaining relevant columns are chosen based on domain expert recommendation. As some clinical assessments have been scattered across different columns due to column name differences between CSV files, we merge these back into a single column per assessment. We also need to generate a unique patient ID by using two encrypted PIN columns and the

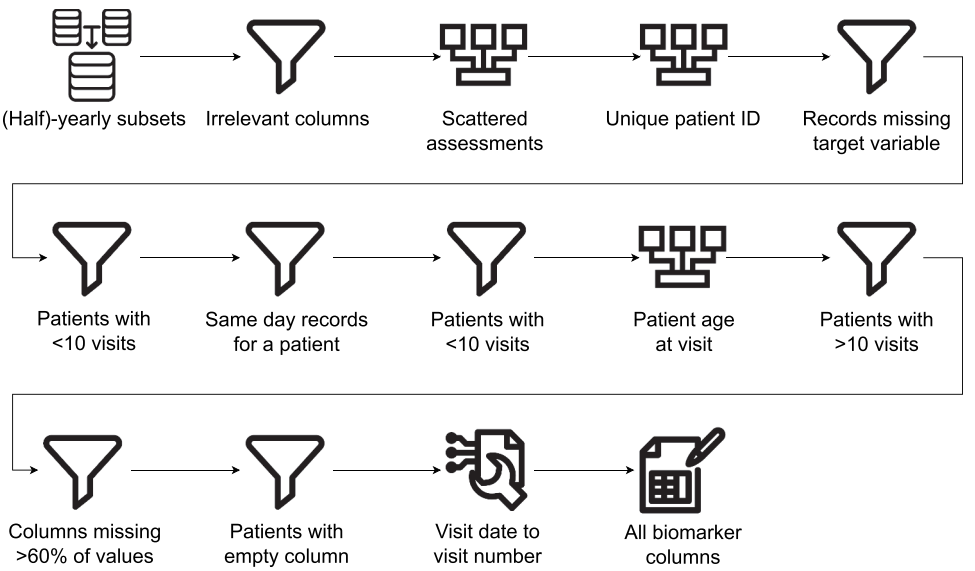


Figure 10. A flowchart of the pipeline's data pre-processing phase for the diabetes dataset. Icons taken from Freepik.

encrypted date of birth of patients to identify them throughout the pipeline. These steps reduce the number of columns to 26.

Next, we remove any records where the target variable is missing and any patients that do not have at least 10 visits recorded in the data. Also, if multiple records were created on a single day, we discard all records on that day except the last one. Because this may have put some patients under the 10 visit threshold, we once again remove any patients that do not have at least 10 visits recorded in the data. This is done twice because of the high computational cost for discarding same-day records. This reduces the dataframe length from 737,010 rows to 86,185 rows, covering 6793 patients. Because we have no information on the age at each visit of patients, we calculate this using the date of birth and visit time. This age is stored in a new column and the date of birth column is dropped. Next, we remove any visits past the tenth visit for each patient, so we have equal length data sequences. This brings the number of rows down to 67,930.

Next, we start handling missing data. First, we remove any columns that are missing over 60% of their values. This removes 12 highly sparse columns, bringing the total down to 14. Next, we remove patients that have a fully empty column across all visits, as we cannot perform forward/backward filling on such columns, and reset the patient IDs. Doing so reduces the number of rows and patients to 66,440 and 6644, respectively. After this, we change the visit dates to visit numbers and finalize the preparation of the data by imputing any remaining missing values using the chosen imputation method.

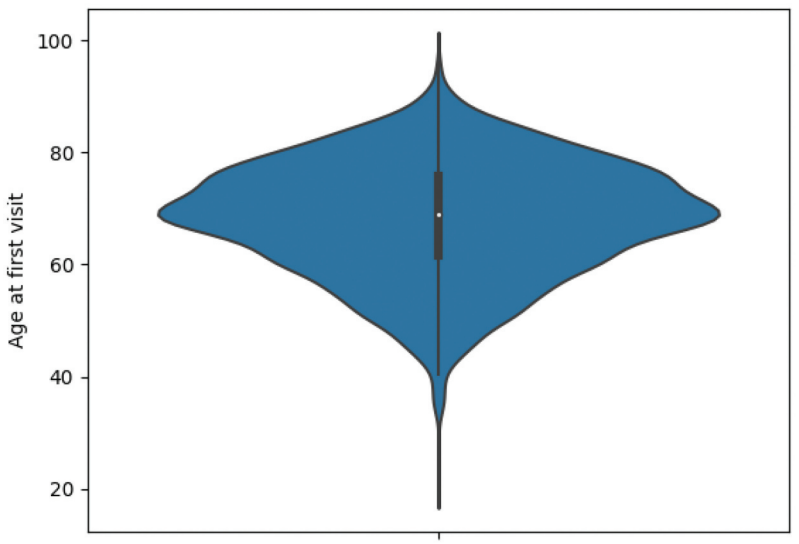


Figure 11. The distribution of patient age at the first visit.

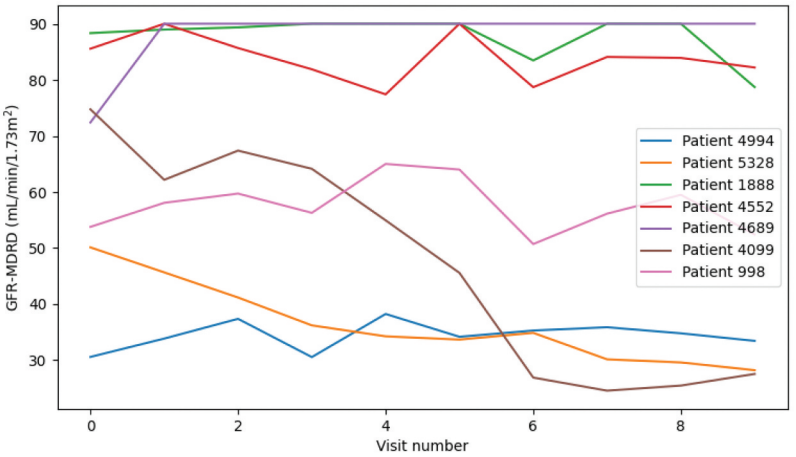


Figure 12. Several examples of patient eGFR assessments over time.

The raw dataset contains 737,010 rows and 861 columns of data, covering 128,793 patients. After completing the data preparation phase, we are left with 66,440 rows and 14 columns of data, covering 6644 patients.

In Figure 11, we show the distribution of patient age after data preparation. Similar to the raw dataset, it is clearly visible that the distribution is skewed heavily toward elderly patients, with little observable change in the age group of 50 years and older. However, when looking at the lower age group, we see that the number of patients decreases drastically during data preparation. Under the age of 30, barely any patients are included. We could not find a clear cause for this change.

A number of examples of what the patient eGFR values look like over time are shown in Figure 12. Similar to the PPMI case study data, we see heterogeneity between patients and high variability in trajectories. At the same time, we also see that the overall trend of most patients is fairly stable. Patient 4099 stands out in that regard, displaying a very strong decrease in eGFR. When comparing these sequences to the raw dataset example sequences, we see a large difference. Many patients in the raw dataset have either very few or no eGFR records, whereas the prepared data contains only patients with a full sequence of 10 visits. We also see a larger range for the eGFR in the prepared dataset, but this is caused by the lack of representative patient sequences in the raw dataset sequence example as a result of data sparsity.

Machine Learning Model

For the diabetes case study, we have the same three hyperparameters to optimize: batch size, learning rate and dropout rate. However, compared to the PPMI case study, we have a much larger dataset. Thus, we choose to increase the batch size to 128, while decreasing the learning rate from the

default 0.001 to 0.0005. In contrast to the PPMI case study, there is no issue with small datasets for this case study, so we do not account as strongly for overfitting to training data that might not be representative of unseen data. Hence, we choose to decrease the dropout rate by 50% to 0.1.

Results

This section is divided into two subsections, each covering the results for one case study. Within these subsections, we first present the results of the comparison between our three imputation methods without the explanation phase. These results are shown for four data splitting seeds and five cross-validated model versions. We do not discuss the predictive performance itself in depth, as development of a best-in-class predictive model is not the aim of this research. After the results of the imputation comparison, we show the output of the pipeline using the best performing imputation method for a number of example patients, including both the prediction and explanation phases.

Case Study 1: PPMI

As described in Section 3, we start by comparing three imputation methods. This experiment is performed on four different data split seeds. In the appendix, Table A1 shows the full results of the experiment. Each seed represents a different data split, and as such, a different distribution of sequences across test and train partitions, as well as cross validation folds.

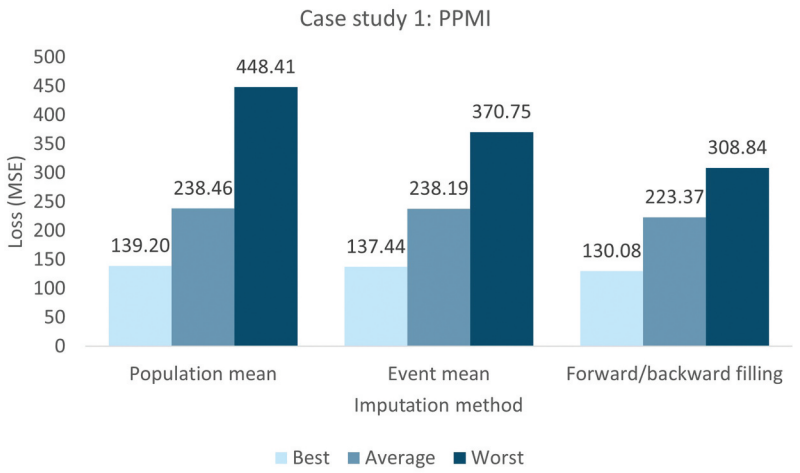


Figure 13. The predictive performance results for the PPMI dataset taken across all cross-validations and data split seeds.

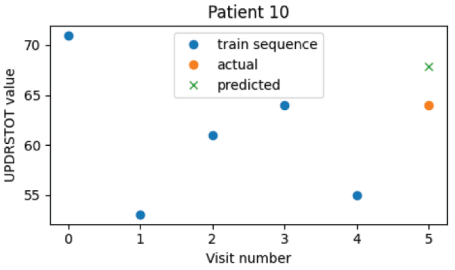
Figure 13 shows a direct comparison between the MSE scores for each imputation method. Across all cross-validations and data split seeds, we show the best performing model, the worst performing model, and the average performance. We show the best and worst model performance to provide an indication of the variability in loss values. From Figure 13, it is apparent that the average loss of the two mean-based imputation methods is almost identical, with event mean imputation performing less than 0.01% better than population mean imputation. Forward/backward filling performs best on average, outperforming both other methods by 6.3%.

We verify the statistical significance of the performance differences by performing a paired t-test, as the samples cannot be considered independent: the majority of the data in the dataset is the same for each trained model, regardless of the data split seed, the cross-validation folds, and the imputation method. In all our comparisons, we test at a significance level of 0.05. This means we are checking if there is a meaningful difference in average performance, with the starting assumption (null hypothesis) being that there is no difference. Applying the test to a comparison between population mean imputation and event mean imputation gives a two-tailed p-value of 0.988, so the hypothesis cannot be rejected. Similarly, the comparison between the population mean technique versus forward/backward filling yields a two-tailed p-value of 0.293, meaning rejection of the null hypothesis fails.

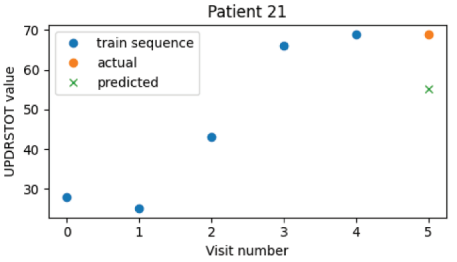
From these paired t-tests, we conclude that neither event mean imputation or forward/backward filling offers a statistically significant performance increase over using population mean imputation. It should be noted that the lack of statistical significance of our observed difference does not mean that the choice of imputation method has no influence on the performance, but that from the results we obtained there is no conclusive evidence that the observed difference in performance is not caused by chance. For the remainder of this case study, we consider forward/backward filling to be the best performing imputation method and use this method for the generation of our further results.

Model Training and Evaluation

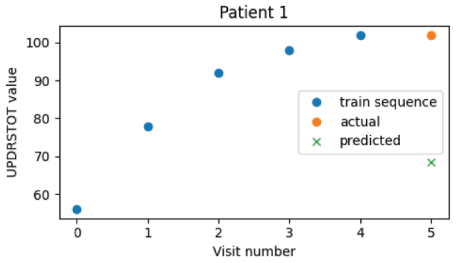
For the generation of the pipeline outputs in this subsection, we use the fold 1 model trained using seed 1998 because this model's loss approximates the average loss for the imputation method we use within 1%. As such, we assume that the chosen model is representative of the pipeline's general performance. In Figure 14, we showcase three example predictions, along with the preceding sequence of the target variable and the ground truth for the predicted value. The first example, patient 10, is an instance of a relatively good prediction, the second (patient 21) an average prediction, and the third (patient 1) a relatively poor prediction.



(a) Patient 10, above average predictive performance.



(b) Patient 21, average predictive performance.



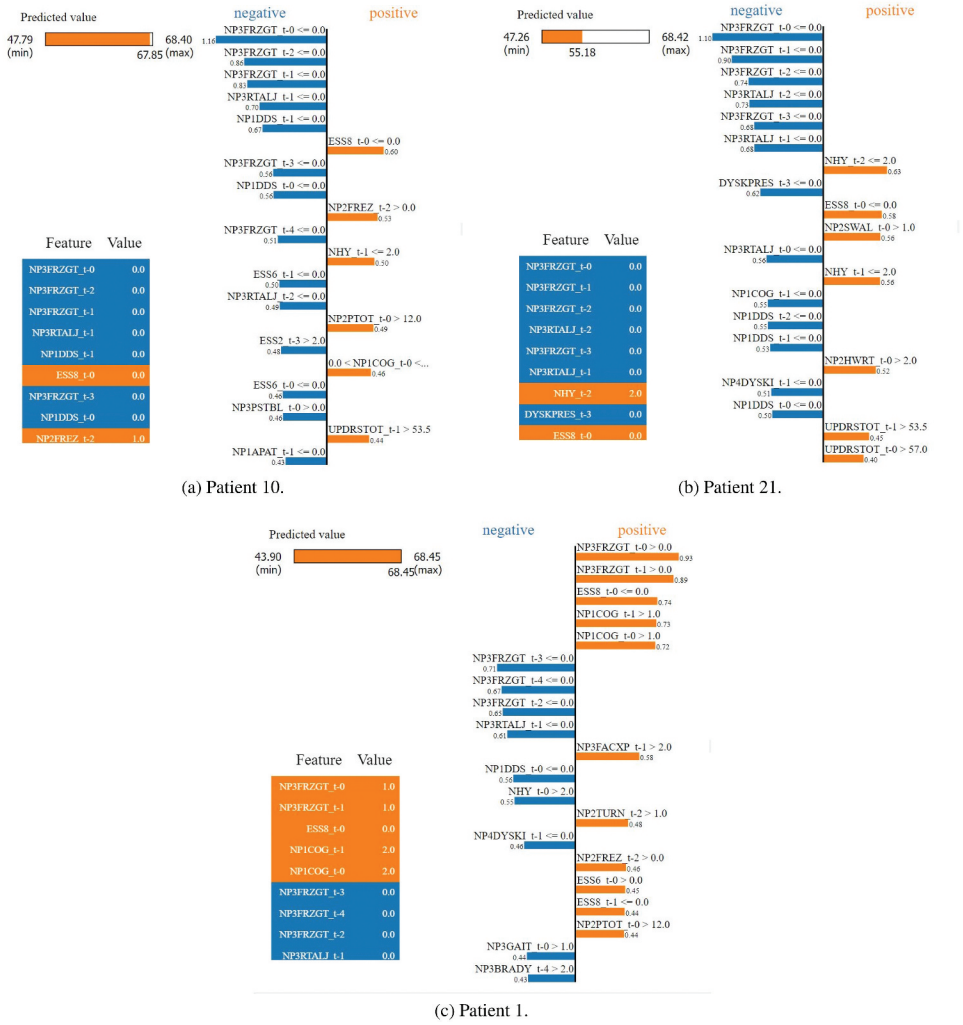
(c) Patient 1, below average predictive performance.

Figure 14. Three plots containing the MDS-UPDRS total score sequence of various patients in the PPMI dataset.

Explainability of Results

When looking for an explanation for the predictions shown, the last phase of the pipeline is of interest. For the same patients we discussed above, the explanations offered by LIME are visible in Figure 15. In each instance, the explanation consists of three elements. The top left element of the explanation, the "Predicted value" bar, shows the range of predictions, as well as the prediction for the instance that is being explained. The rightmost element contains a discretized view of the top 20 features, in descending order of importance or influence. Negative influence (depicted in blue) means that the predicted value is decreased by the feature, positive influence (depicted in orange) means an increase in predicted value. Each feature name consists of an assessment and the timepoint at which the assessment was done. On the bottom left, a table showing the top 20 features is included. This table contains the feature name, as well as the exact value of that feature at that timepoint, ranked by descending feature importance. In the environment in which the explanations are generated and presented, this table can be scrolled through to show all 20 features. In the following paragraphs, we list our main observations and takeaways from the explainability graphs.

One peculiar observation from the explanations as shown in Figure 15 is the marginal influence of features. Furthermore, looking at Figure 15 plot a) specifically, we see in the leftmost element that the predicted value is roughly the maximum possible prediction, but the five most important features in the



(a) Patient 10.

(b) Patient 21.

(c) Patient 1.

Figure 15. Three plots containing the LIME explanation for the progression prediction of various patients in the PPMI dataset.

middle element are shown to have a negative effect on the prediction. As such, the explanation points toward a relatively low prediction value, while the actual prediction value is very high. It is possible that this is a consequence of the low differences in influence between the various features and time-points. As the influence of each feature is small, it could be that outside the top 20 features shown, there are relatively many features that have a positive influence that bring the total prediction value up.

Impact of Previous Measurements. An intriguing finding is that in all the examples we looked at, the earlier measurements of the target variable do not appear among the top 15 most important features. Rather, what we find is

a range of very specific motor scores (UPDRS) that are considered to have the greatest impact. In the explanations for the prediction of both patients 10 and 21, depicted in plots (a) and (b) of Figure 15 respectively, the UPDRS total score is ranked nineteenth among the most important features. In the explanation for patient 1, shown in Figure 15 (c), the UPDRS total score is not included in the top 20 features at all. This is striking, because we would expect the progressive nature of Parkinson's disease to lead to a strong influence of previous UPDRS total scores on the predictions. Looking at the correlation between various features in the prepared data and the UPDRS total score shown in Appendix B (Figure B1), we see that by far the strongest correlations exist between the UPDRS total score and the partition total scores NP2PTOT and NP3TOT. This is logical, as the UPDRS total score is a summation of all partition total scores. However, similar to the UPDRS total score, the partition total scores are barely, if at all, included in the explanations. Only Figure 15 plot (c), shows the NP2PTOT score in the top 20 most important features, at rank 18.

Inadequate Range of Prediction. A different remarkable finding from the explanations, is that the maximum MDS-UPDRS total score that can be predicted is reported to be just under 70, according to LIME. However, when looking at the sequence for patient 1 in Figure 14 plot (c), we see that this is not an adequate range. Patient 1 has only a single visit during which the score was below that boundary. Similarly, in Figure 14 plot (a), we see that patient 10 has a score above 70 for their baseline visit as well. As such, it appears that the predictive model is not able to account for the full range in which the target variable exists.

Domain Specific Validation

As part of the validation of the results by a domain expert, there are several points to be noted here. Looking at the result from a clinician's perspective, the predictive output throughout the patient group shows a tendency to underpredict the UPDRS much more often than overpredict (e.g. Patient 1). In general, these predictions would – as of now – serve little purpose in a clinical setting.

Focusing on the explanations given for the predictions, there are some obvious inconsistencies. Any item scored with 0 (absence of a sign or symptom) should have a diminishing influence on the total UPDRS prediction (contrary to ESS8 in all examples of Figure 15). Likewise, a higher score should lead to an increase in predicted UPDRS, though this is not consistently observed. In addition to this, it appears that highly specific, singular items are attributed a lot of weight, especially since some of the high-ranking factors are not necessarily due to only PD (e.g. mental health, incontinence).

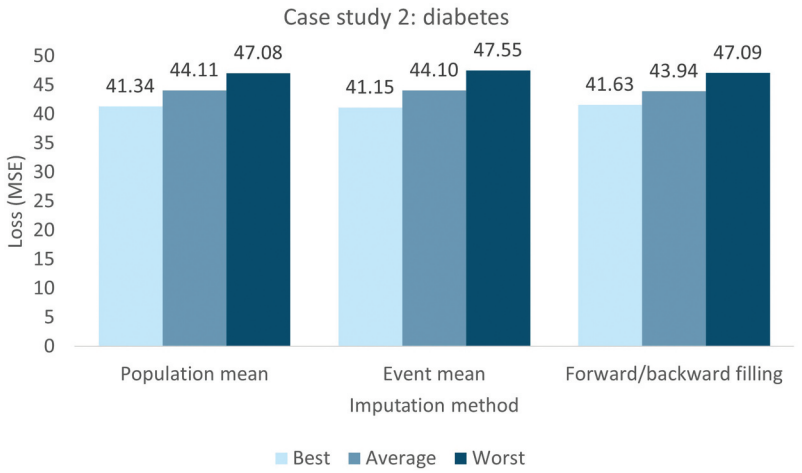


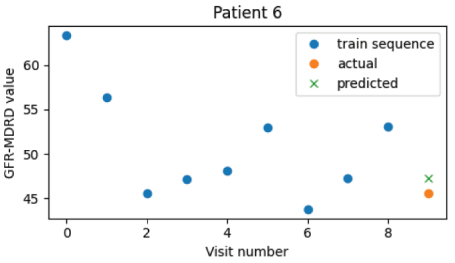
Figure 16. The predictive performance results for the diabetes dataset, taken across all cross-validations and data split seeds.

Case Study 2: Diabetes

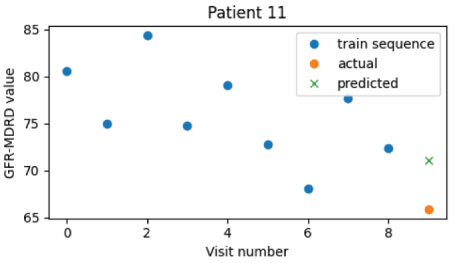
Again, we start by comparing our three imputation methods. Table A2 in the appendix contains the full results of this comparison. Figure 16 shows a direct comparison between the best, worst and average performance across all cross-validations and data split seeds to provide an indication of the variability in loss values. Once more, it is clearly visible that the average loss of the two mean-based imputation methods is almost identical, with event mean imputation performing less than 0.01% better than population mean imputation. Forward/backward filling performs best on average, but the difference in performance for this method is less than 0.01% compared to the other two methods as well.

Effect of Imputation Techniques

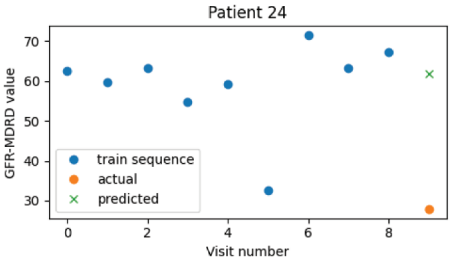
Similar to the PPMI results, we apply the Paired t-test to determine whether any of the observed performance differences are statistically significant. For the first comparison, population mean imputation versus event mean imputation, we obtain a two-tailed p-value of 0.924. Hence, the null hypothesis cannot be rejected at a significance level of 0.05. The second comparison, population mean imputation and forward/backward filling, gives a two-tailed p-value of 0.539. This exceeds the significance level of 0.05 and therefore fails to reject the hypothesis. From these test results, we conclude that no tested imputation technique offers a statistically significant performance increase over population mean imputation. However, because forward/backward filling offers the lowest MSE in the comparison, we consider that technique to be the best performing imputation method and use this method for the generation of our further results in this section, as we did for the other case study.



(a) Patient 6, above average predictive performance.



(b) Patient 11, average predictive performance.



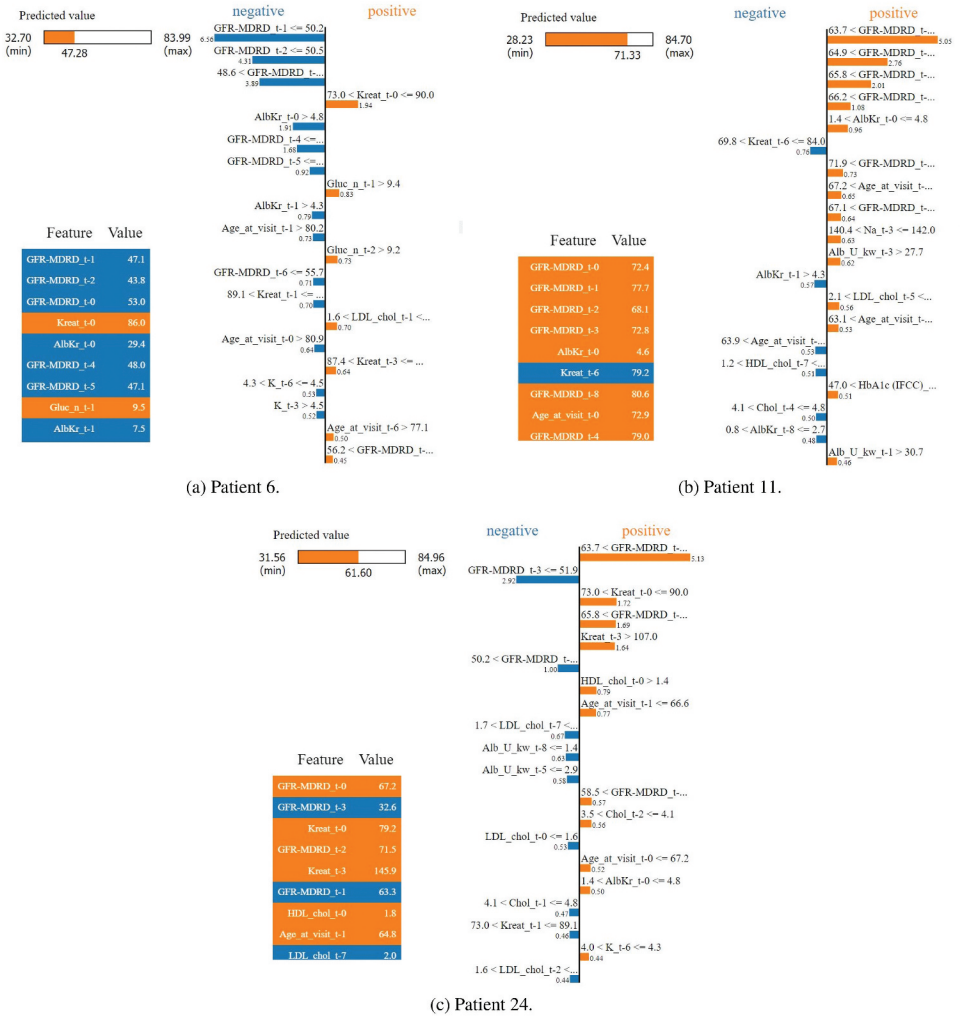
(c) Patient 24, below average predictive performance.

Figure 17. Three plots containing the GFR-MDRD (in mL/min/1.73 m2) sequence of various patients in the diabetes dataset.

Model Training and Evaluation

For the generation of the pipeline outputs in this subsection, we use the fold 5 model trained using seed 25 because this model's loss approximates the average loss for the imputation method we use within 1%. As such, we assume the chosen model is representative of the pipeline's general performance. In Figure 17, we showcase three example predictions, along with the preceding sequence of the target variable and the ground truth for the predicted value. The first example, patient 6, is an instance of a relatively good prediction, the second (patient 11) an average prediction, and the third (patient 24) a relatively poor prediction. In plot (a) of the figure, it is visible that while the GFR-MDRD values have an erratic trend, there are no excessively large changes between visits. The biggest jump is around 10 mL/min/1.73 m2, between visits 1 and 2 (counting from 0). The predictive model predicts a slightly too high GFR-MDRD, but the error is less than 2 mL/min/1.73 m2, which indicates a prediction that is easily within the 15% (in this case roughly 6.9 mL/min/1.73 m2) biological margin that is used in clinical settings.

(a) Patient 6, above average predictive performance.
(b) Patient 11, average predictive performance
(c) Patient 24, below average predictive performance.



(a) Patient 6.

(b) Patient 11.

(c) Patient 24.

Figure 18. Three plots containing the LIME explanation for the progression prediction of various patients in the diabetes dataset.

Explainability of Results

The explanations for patients 6, 11, and 24 are shown in Figure 18. Once more, the top left element of the explanation shows the range in which predictions can fall and the prediction value. The rightmost element contains a discretized view of the top 20 features and their influence on the prediction value, the bottom left element contains the exact value for each of the included features.

Significance of Short Lag. When looking at the explanations for the diabetes case study, we immediately notice that the previous values of the target variable are strongly represented in the explanations. For each explanation shown, many of the most recent eGFR values are highly influential on the final prediction. Mixed between these, however, we also observe values such as the

Kreat (creatinine) biomarker and blood glucose exercising influence on the final prediction.

It is consistent with the method with which eGFR is computed in practice. eGFR or GFR is calculated using different formulas from creatinine e.g. MDRD: creatinine, age, gender and skin type. So only creatinine is the determinant of GFR in an individual patient so they are surrogates for each other.

Furthermore, we see that for some specific assessments, a feature that is multiple visits ago is included in the top 20 features. This suggests that the specific feature stands out amongst other assessments taken in the same visit, and that the predictive model recognizes such data points as characteristic for the patient.

Domain Specific Validation

From the domain perspective, the first observation is that the degree to which the model can predict kidney function seems adequate, with 84.1% of all predictions being within the commonly used 15% analytical margin around the ground truth eGFR. This level of performance could give the laboratory the possibility to start classifying patients in a low and high risk category for decline in kidney function. Based on such classification, low-risk patients could have a lowered monitoring frequency for kidney function in the future.

The explanations for the three example patients, shown in Figure 18, make sense in general. This is as expected because the available assessments are directly or indirectly related to the eGFR. One *surprising element* was the absence of blood glucose among the most influential features, which they only observed for the above average performing prediction. As such, it is possible that this relation is not captured very well by the prediction model.

One general remark from the domain expert perspective, were the relatively high knowledge requirements to understand the provided explanations. We took this comment into consideration while identifying further research opportunities and found that this issue has also been raised by Dieber and Kirrane Dieber and Kirrane (2020), albeit in a different problem context. This is discussed further in Section 5.

Discussion

In this section, first various limitations of the research are discussed. Subsequently, a number of recommendations for future research are provided.

Data Imputation

Both case studies have missing values in rows and columns. Although it is not advisable to use imputation to handle missing data, the available alternatives are scarce and limited in their utility for real-world datasets. The two main alternatives mentioned in the literature are record removal or the use of ML models that are robust to missing data. The applicability of both is deemed infeasible due to the reasons outlined below.

The small size of the PPMI dataset combined with the degree of data sparsity prevents us from using record removal as an alternative to imputation. Following this approach leads to an empty dataset after pre-processing to suit the prediction model input requirements. For the second alternative, we identifies only one ML model that meets the criteria we have set (time-series input data, continuous prediction value, RNN-based model) whilst also being robust to sparse input data: a GRU-based model proposed by De Brouwer et al. (2021). However, the highly complex nature of the model made the integration between the predictive model and the Python implementation of the LIME framework unfeasible.

For these reasons, we were forced to use imputation to address the missing data problem in our ML pipeline. To mitigate the drawbacks of imputation, we compared techniques, performed cross-validation, and repeated experiments for multiple seeds.

Data Imputation Effects on Explanations

A key challenge of data imputation is how it affects the steps following the prediction of disease progression in our process. The imputed values can compromise the reliability of the generated explanations.

LIME functions by creating a simple model that represents what is happening locally around a specific prediction. Just like the prediction model, LIME is also not robust to the imputed data. The imputed values could influence the prediction performance as well as the explanation generated by LIME. LIME could very well based its decision for a feature being influential due to the imputed values.

However, this does not necessarily mean such an explanation is incorrect. In fact, if LIME emphasizes a feature as a result of an imputed value, the real-world feature importance is not reflected, but the explanation does reflect the feature importance within the model. This distinction is noteworthy, because it highlights the ability of the explanation phase to reveal shortcomings of the predictive phase.

Imputation Comparison

Further considerations were necessary as a result of the choice of imputation methods. As described in Section 3, we use three imputation methods. Population mean imputation and event mean imputation can fill missing values as long as any other patient record exists for that measurement. This is highly probable, as the population in the datasets is in the hundreds or thousands. Forward/backward filling, however, requires existing values for a specific patient. In other words, only a number of records equal to the sequence length can be used, a number that is orders of magnitude smaller. As such, the likelihood of a sequence having a column for which imputation cannot be performed due to the absence of values increases drastically.

If certain columns cannot be filled by forward/backward filling, removing the column is not the preferable solution. Since the majority of other sequences are unlikely to face this issue, removing the column would result in the unnecessary loss of far more information than merely excluding the problematic sequence. Hence, we choose the latter approach. While this methodology solves the problems with the imputation step itself, it can be considered undesirable for our comparison of imputation methods. After all, removing any sequences with empty columns to accommodate forward and backward filling effectively "hobbles" the other imputation techniques. These techniques could include the excluded sequences and information without difficulty, thus possibly improving predictive performance.

We considered doing an additional comparison between using all sequences and using the sequences suited to forward and backward filling, but ultimately decided against this. This decision is justified by noting that the data distribution across data splits and cross-validation folds would be completely different regardless of the set seeds, due to the input data being different. There is no way of fairly comparing only the effect of having additional sequences, because there would also be an effect on the performance caused by the different distribution of data across the various splits and folds.

Furthermore, any unobserved performance difference due to this limitation must also have had an impact large enough to be statistically significant. Considering that the differences in results mentioned in Section 4 are far from being statistically significant, it is unlikely that the possible increase in training data would have led to a significant performance gain for the mean-based imputation methods. In light of this, we must assume that the effect of choosing a certain imputation method on the applicability in a clinical setting is negligible. To impact performance positively, emphasis should instead be on mitigating the amount of data for which imputation is required.

Data Dimensionality

As observed in Section 4, in the context of the PPMI case study the explanation results do not correlate with our and the domain expert's expectations, whereas the explanation results for the diabetes case study do. We hypothesize that the observed phenomena in the PPMI case have the same underlying cause. We note the following: Rad et al. (2021) state that as the dimensionality of data increases, the likelihood of feature values incidentally being correlated with events or anomalies increases as well. They explicitly add that for time-series data, high dimensionality is even more problematic. As time-series data does not only contain x amount of features, but also y timepoints at which each feature exists, the "effective" dimensionality of the data increases to $x * y$. As such, the likelihood of incidental correlation is high. If this knowledge is applied to the PPMI dataset, we find that the dataset effectively contains 483 features over all timepoints (83 feature columns * 6 visits). That number is almost twice the number of sequences available for training the pipeline, which is 247 after removing the test set.

Considering the extremely high effective dimensionality of the PPMI dataset, it is then logical that the predictive performance in this case study is relatively lower than the predictive performance for the diabetes case study. In turn, we expect the explanations to reflect a poor prediction model. However, explanations reflecting a subpar prediction model may not be the only issue resulting from the dimensionality problem. Because the LIME framework is based on local approximation, as discussed in Section 2, the high dimensionality of the data also leads to poor fitment of the local approximation in the LIME framework. Molnar (2020) mentions that this dimensionality problem is exacerbated by the calculation of the kernel width in the Python implementation of LIME. The kernel width is what determines how close perturbed instances must be to influence the model. The larger the kernel, the farther away an instance can be while still exercising influence on the approximation model. Molnar explicitly mentions that the Python implementation of LIME calculates the kernel width using Equation 1.

$$kernel_width = 0.75 * sqrt(n_{columns}) \tag{1}$$

If we calculate the kernel width for the diabetes dataset and the PPMI dataset using the same equation, we see that the diabetes dataset has a kernel size of 8, as shown in Equation 2.

$$kernel_width = 0.75 * sqrt(120) \approx 8 \tag{2}$$

whereas the PPMI dataset has a kernel size of 16, as shown in Equation 3.

$$kernel_width = 0.75 * sqrt(483) \approx 16 \tag{3}$$

Thus, even though the PPMI dataset contains 23 times less data sequences, the kernel size for this dataset is twice the kernel size for the diabetes dataset. This oversize kernel likely causes too many perturbed instances to have an impact on the local approximation, leading to the poor explanations observed in the results for the PPMI case study.

Generalisability of the Pipeline

A high degree of generalizability is a desirable feature for most systems, and this is no different for our pipeline. To assess the degree to which our proposed pipeline is generalizable, we use a train-test split as well as cross-validation. There are two primary reasons why out-of-sample performance may be sub-par. The first is that the pipeline is simply not generalizable, and only suits a specific case due to the architecture of the pipeline. The second is that a data distribution problem is present, meaning that the model is not being trained on data that is representative of the population.

In the PPMI case study, we see extreme variability in the predictive performance of the pipeline as shown in Table A.1. Across different seeds for splitting the data as well as cross-validation model variants, there is a very high standard deviation in the MSE. In the worst case for example, seed 1998, we see a standard deviation of 37.4% of the average MSE when using population mean imputation. For seed 787 on the other hand, the same imputation method results in a much lower standard deviation of 9.0% of the average MSE. This suggests that the data distribution over the splits and folds has a very strong effect on the predictive performance.

One of the most logical explanations for the high variability in this context would be the size of the PPMI dataset. If we look at the difference in size between the diabetes and PPMI datasets, we see that once processed, the diabetes dataset contains nearly 6700 patients, whereas PPMI only contains 290. As the train-test split is 85–15%, this only gives us a test set of 43 patients. Imbalanced data distribution and biological heterogeneity have a stronger influence on smaller datasets. To further solidify this explanation, the standard deviation across cross-validation within the diabetes dataset is significantly lower than the PPMI dataset. The larger dataset size appears to correlate with a lower degree of variability.

Temporal Irregularity Assumptions

In this research, we do not actively account for temporal irregularity. We note that the problem of temporal irregularity exists in (healthcare) time-series data, but we do not actively align the time of data points. As briefly discussed in Section 3, this assumption is not without reason.

For our first case study, PPMI, we justify the assumption that any temporal irregularity does not require intervention because of the structured planning that is the backbone of the PPMI study. The visits we use in our research are yearly, and each visit is registered with the appropriate event identifier. As such, even if patient *A* has their second visit exactly a year after their first visit, and patient *B* has their second visit 13 months after their first visit, the influence of this different interval on the captured trend of the progression should be minimal.

For our second case study, diabetes, there is no such strict planning. However, according to the domain expert for this case, we can base our assumption on the Nederlands Huisartsen Genootschap (NHG) *(English: Dutch General Practitioners Association)* guidelines for monitoring diabetes patients (Barents et al. 2023). Because the data in the dataset is generated by primary care providers such as general practitioners, this guideline is followed. Hence, the data does not include (parts of) sequences which are recorded within only a few days as a result of constant monitoring in the hospital. The reason for the request of the assessments is important for justifying our temporal irregularity assumption too, because the dataset mostly contains visits where the primary goal of the visit was monitoring the HbA1c biomarker, as per the NHG guideline mentioned above. As the HbA1c biomarker is an indication of blood glucose levels over the past 3 months, this is the interval that the guideline suggests between visits. Thus, it is likely that all measurements are taken at such intervals, along with the HbA1c measurement. As a result, the likelihood of semi-consistent intervals is high enough to assume minimal impact of any irregularity across the entire dataset population.

Predictive Model

The predictive model we use in this research is constructed using existing, state-of-the-art components. Because some of these components are – as of yet – not widely used in the context relevant for this research, the flexibility of some implementations is not very high. All of the recommendations in this subsection are related to missing data, so it is possible that all of them require addressing simultaneously from an implementation perspective.

An example of flexibility that would be beneficial for the performance of the pipeline as a whole, is the ability for the predictive model to handle ragged tensors. For example, three patients that have six, five, and six visits respectively. As currently implemented, the model in the pipeline cannot handle this and would require the patients with six visits be pruned back to five visits, or for the patient with five visits to be removed, leaving only two patients. In either case, information is lost as a result of inflexibility of the model. Hence, we recommend that the flexibility of the model's input is to be improved.

While the form of flexibility described above can be seen as revolving around missing visits at the end of a sequence, the next flexibility improvement concerns missing visits within a sequence. Such flexibility works differently for the two case studies in this research: in the PPMI dataset, visit records are explicitly connected to a planning. As such, a patient's second visit may be labeled "visit 3" rather than "visit 2," because the visit takes place at the intended time for "visit 3." The record "visit 2" remains non-existent for that patient. In the diabetes dataset, there is no such planning. Each visit simply follows up on the last registered visit and will be numbered as the direct followup. However, if the model is capable of handling a missing visit, flexibility regarding such instances is increased dramatically for both case studies. In the PPMI case, a patient visit could simply be considered missing if a later visit is present for the patient. In the diabetes case, it could be possible to use the time interval between visits to determine whether a visit in between two visits is "logically" missing. For either case, such flexibility would improve the amount and correctness of information captured in the dataset, subsequently improving predictive performance.

Finally, we note the lack of flexibility regarding missing values within a single record. The current implementation of the pipeline requires the input data to be complete, without a single missing value. As discussed earlier as a limitation in this section, due to technical feasibility, we were forced to use imputation regardless of the known associated drawbacks. Therefore, we recommend researching whether the predictive model could be made robust to missing values without imputation, and comparing whether this leads to improved predictive performance.

Explanations

There are two aspects to the improvement of the explanations given by our pipeline. The first is the fidelity or correctness of the explanations when compared to the ground truth, the second is in the presentation of the explanations.

As mentioned in an earlier part of this chapter, we find that for the PPMI dataset, there is most likely a dimensionality problem leading to very poor explanations. In the explanation of the problem, we mentioned the calculation of kernel width for the Python LIME implementation. This calculation might be a solution to the high dimensionality problem for LIME. If the calculation for the kernel width is changed so that it remains lower for high dimensional data, the explanations for this data may better reflect the true inner workings of a prediction model. Thus, we recommend that the influence of changing the kernel width calculation is tested in further research.

In our pipeline, we use the LIME framework to explain our predictions. LIME is an established framework and is used as a benchmark for other

explainability frameworks on a number of occasions as mentioned by Dieber and Kirrane (2020). However, they also mention that LIME is not properly bench-marked itself. Dieber and Kirrane perform an evaluation of LIME from a usability perspective, but they do not test the fidelity of the framework. We define fidelity as the degree to which the explanation for a prediction generated by a framework matches the ground truth of why the prediction should be what it is. Because there is very little, if any testing done to confirm the fidelity of LIME, the fidelity of the framework is to some extent uncertain. In this research, we do not perform formal validation either, instead only using anecdotal evidence from domain experts for our two case studies. As a suggestion for future research, it would be valuable to assess the performance of LIME including fidelity. A possible method for this would be to use the Co-12 properties and associated evaluation methods introduced by Nauta et al. (2023).

As a final recommendation, we would suggest improving the presentation of the explanations provided by LIME. The paper by Dieber and Kirrane (2020), concludes that the visualization of explanations offered by LIME are not very intuitive to interpret without background knowledge of what each presented element of the explanation represents. Furthermore, some parts of the explanations are cut off when the explanation is presented. These cutoffs lead to, for example, discretization boundaries being unreadable. The raw information of the explanation, such as feature influence on the prediction value, is accessible in the Python LIME implementation. As such, it is possible to use this information to create a custom, improved presentation of the explanation. Examples of such an improvement are a clearer guide of what each element of the explanation represents, as well as a different display of the feature influence plot.

Clinical Implications

The need for and importance of transparent machine learning algorithms in medicine cannot be understated. They not only support prognostication and clinical management but could also help to identify new clinico-biological features relevant to disease courses. If these predictive models function on a patient-specific basis, they furthermore harbor a great potential for personalized medicine. Since the model can update its predictions based on newly available data, it is also capable of dynamic prognostication as opposed to most prognostication in medicine that is solely based on the state at initial diagnosis.

Parkinson's disease (PD) is a neurodegenerative disease with highly variable disease courses. Accurate prediction of the disease progression is very challenging. Several clinical experts in the field state, that a reliable explainable AI approach could aid in medical decisions and guide therapeutic adjustments (e.g. invasive therapies such as deep brain stimulation). If a disease-modifying

(neuroprotective) therapy for PD becomes available in the future, it could assist in identifying patients at the highest risk. Moreover, it could help decrease uncertainty in both the patient and the clinicians leading to better acceptance and patient-doctor-relationships.

Taking the results into consideration, an important aspect of this research is its focus on the UPDRS as its sole target variable, a clinical score that incorporates the rater's and the patient's observations. As such, it is highly susceptible to non-PD-related influences like comorbidities, socioeconomic environment, and personality. Even though PD can be considered a chronic and slowly progressing disease, this does not necessarily reflect in the clinical symptoms captured by the UPDRS, that does not progress in a linear way. Other temporary and permanent diseases as well as changes in treatment (still purely symptomatic) can have drastic and hardly foreseeable effects on the UPDRS in either direction. Another limitation and possible vulnerability to error of the UPDRS as a proxy for disease severity is its suboptimal inter-rater reliability (meaning deviations in assessments by different clinicians).

As a potential for further research regarding the PD case study, reducing the number of prediction factors and as such the data dimensionality by grouping items into the four parts of the UPDRS and potentially including other biological or clinical factors could improve results. Additionally, predicting over larger periods of time could decrease the noise through fluctuations of the UPDRS. While there is no solid biomarker to measure disease progression, another target variable might be a better surrogate of the state of PD (e.g. Hoehn and Yahr scale see e.g. Bhidayasiri and Tarsy 2012). With respect to the second case, diabetic nephropathy, which is highly prevalent in diabetes mellitus, is associated with an increased risk of cardiovascular disease, end-stage kidney failure, and premature death. Several experts in diabetic care and management have shared their insights on the possibility of predicting the development of diabetic nephropathy based on patient history. They describe this approach as having added value, as it can personalize treatment plans in high-risk patients, e.g., by providing dietary advice and/or avoiding nephrotoxic medication, and thereby potentially preventing or delaying kidney failure and its related adverse effects. Moreover, it may also decrease the burden on the healthcare system since it may also predict patients with a low risk of developing kidney failure and thereby potentially reduce the frequency of visits to the healthcare providers and routine checkups.

Conclusion

In this research, we design and develop an end-to-end machine learning pipeline. The pipeline addresses the research problem : state-of-the-art machine learning models for prediction of disease progression are not adequately transparent or interpretable. Our pipeline does offer

transparency and interpretability, whilst using a state-of-the-art prediction model architecture. The demonstration of the functionality of the pipeline, through applying it to two diverse case studies with very different datasets, shows that the pipeline is applicable to a range of diseases, even if particular components do not perform well in their current form. Furthermore, a domain expert and data owner for one of the case studies mentions that the predictions and explanations provided by our pipeline align with their expectations, and that the majority of explanations offered by the pipeline appear logical. For the other case study, we observe poor (explanatory) performance and hypothesize that this is due to the extremely high dimensionality of the dataset. We also provide possible solutions for the poor performance in case our hypothesis is correct.

This research identifies a number of opportunities for future work. These opportunities are primarily focused on two aspects of the pipeline: the predictive model and the explanations. For the first aspect, the predictive phase, we recommend improving the flexibility of the predictive model with regard to input data. Valuable improvements include the ability to handle data sequences of unequal length within a dataset and robustness to missing records or missing values within a record. These improvements may increase the number of disease contexts that the pipeline can be applied in. We provide two recommendations related to the explanation aspect of the pipeline. We suggest investigating whether changing the kernel width calculation for LIME leads to improvements in the fidelity of generated explanations. Such changes could prevent fitment problems with the local approximation model that LIME trains as a result of an oversize kernel width. Furthermore, we recommend improving the visualization of the explanations offered by LIME, as both literature and a domain expert in this research consider the visualizations difficult to interpret.

As a potential for further research regarding the PD case study, reducing the number of prediction factors and as such the data dimensionality by grouping items into the four parts of the UPDRS and potentially including other biological or clinical factors could improve results. Additionally, predicting over larger periods of time could decrease the noise through fluctuations of the UPDRS. While there is no solid biomarker to measure disease progression, another target variable might be a better surrogate of the state of PD (e.g. Hoehn and Yahr scale see e.g. Bhidayasiri and Tarsy 2012).

There is growing evidence that machine learning systems may support or even outperform humans in certain medical tasks. However, acceptance and Implementation in clinical practice is still limited, in part due to ethical and medico-legal concerns. Explainable AI has the potential to help overcome some of these barriers, enhance transparency and lead to better clinical decision-making and patient outcomes. This paper contributes to this endeavor.

Notes

1. https://github.com/marcotcr/lime.
2. https://github.com/philipperemy/keras-attention.
3. https://github.com/StijnBerendse/MSc_thesis.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

ORCID

Faizan Ahmed http://orcid.org/0000-0002-2760-6892

Data Availability Statement

We used two datasets for our research: PPMI and the diabetic data. Access to these datasets is provided for analysis purposes only, and due to restrictions from the data owners, we cannot provide direct access to these datasets. However, the code used to process this data is accessible. All notebooks and scripts used or written for this research can be found on GitHub https://github.com/StijnBerendse/MSc_thesis.

References

Alicic, R., M. Rooney, and K. Tuttle. 2017. Diabetic kidney disease: Challenges, progress, and possibilities. *Clinical Journal of the American Society of Nephrology* 12 (12):2032–45. doi: 10.2215/CJN.11491116.

An, Y., K. Tang, and J. Wang. 2021. Time-aware multi-type data fusion representation learning framework for risk prediction of cardiovascular diseases. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*. doi: 10.1109/TCBB.2021.3118418.

Association, American Diabetes. 2013. Diagnosis and classification of diabetes mellitus. *Diabetes Care* 37 (Supplement_1):S81–90. ISSN: 0149-5992. doi: 10.2337/dc14-S081.

Barents, E. S. E., H. J. G. Bilo, M. Bouma, M. Dankers, A. De Rooij, H. E. Hart, S. T. Houweling, R. G. IJzerman, P. G. H. Janssen, A. Kerssen, et al. 2023. Diabetes mellitus type 2. https://richtlijnen.nhg.org/standaarden/diabetes-mellitus-type-2#samenvatting-controles.

Bhagwat, N., J. D. Viviano, A. N. Voineskos, M. M. Chakravarty, and Initiative Alzheimer's Disease Neuroimaging. 2018. Modeling and prediction of clinical symptom trajectories in Alzheimer's disease using longitudinal data. *PLOS Computational Biology* 14 (9):e1006376. doi: 10.1371/journal.pcbi.1006376.

Bhidayasiri, R., and D. Tarsy. 2012. Parkinson's Disease: Hoehn and Yahr Scale. In Movement disorders: A video atlas: A video atlas, 4–5. Totowa, NJ: Humana Press.

Cervantes, J., F. García-Lamont, L. Rodríguez, and A. Lopez-Chau. 2020. A comprehensive survey on support vector machine classification: Applications, challenges and trends. *Neurocomputing* 408 (May):189–215. doi: 10.1016/j.neucom.2019.10.118.

Chan, L., G. N. Nadkarni, F. Fleming, J. R. McCullough, P. Connolly, G. Mosoyan, F. El Salem, M. W. Kattan, J. A. Vassalotti, B. Murphy, et al. 2021. Derivation and validation of a machine

learning risk score using biomarker and electronic patient data to predict progression of diabetic kidney disease. *Diabetologia* 64 (7):1504–15. doi: 10.1007/s00125-021-05444-0.

Chen, L., A. J. Saykin, B. Yao, F. Zhao, and Initiative Alzheimer's Disease Neuroimaging. 2022. Multi-task deep autoencoder to predict Alzheimer's disease progression using temporal DNA methylation data in peripheral blood. *Computational and Structural Biotechnology Journal* 20:5761–74. doi: 10.1016/j.csbj.2022.10.016.

Chen, X., W. Gao, J. Li, D. You, Z. Yu, M. Zhang, F. Shao, Y. Wei, R. Zhang, T. Lange, et al. 2021. A predictive paradigm for COVID-19 prognosis based on the longitudinal measure of biomarkers. *Briefings in Bioinformatics* 22 (6). doi: 10.1093/bib/bbab206.

Cox, D. R. 1972. Regression models and Life-Tables. *Journal of the Royal Statistical Society: Series B (Method- Ological)* 34 (2):187–220. ISSN: 00359246. Accessed March 31, 2023. doi: 10.1111/j.2517-6161.1972.tb00899.x.

Dadu, A., V. Satone, R. Kaur, S. Hadi Hashemi, H. Leonard, H. Iwaki, M. B. Makarious, K. J. Billingsley, S. Bandres-Ciga, L. J. Sargent, et al. 2022. Identification and prediction of Parkinson's disease subtypes and progression using machine learning in two cohorts. *Npj Parkinson's Disease* 8 (1):172. doi: 10.1038/s41531-022-00439-z.

Dang, T., J. Han, T. Xia, D. Spathis, E. Bondareva, C. Siegele-Brown, J. Chauhan, A. Grammenos, A. Hasthanasomba, R. A. Floto, et al. 2022. Exploring longitudinal cough, breath, and voice data for COVID-19 progression prediction via sequential deep learning: Model development and validation. *Journal of Medical Internet Research* 24 (6):e37004. doi: 10.2196/37004.

De Brouwer, E., T. Becker, Y. Moreau, E. K. Havrdova, M. Trojano, S. Eichau, S. Ozakbas, M. Onofrj, P. Grammond, J. Kuhle, et al. 2021. Longitudinal machine learning modeling of MS patient trajectories improves predictions of disability progression. *Computer Methods and Programs in Biomedicine* 208:106180. doi: 10.1016/j.cmpb.2021.106180.

Dieber, J., and S. Kirrane. 2020. Why model why? Assessing the strengths and limitationsofLIME. *arXiv:2012.00093 [cs.LG]*. doi: 10.48550/arXiv.2012.00093.

Dinov, I. D., B. Heavner, M. Tang, G. Glusman, K. Chard, M. Darcy, R. Madduri, J. Pa, C. Spino, C. Kesselman, et al. 2016. Predictive big data analytics: A study of Parkinson's disease using large, complex, heterogeneous, incongruent, multi-source and incomplete observations. *PLOS ONE* 11 (8):e0157077. doi: 10.1371/journal.pone.0157077.

El-Sappagh, S., H. Saleh, F. Ali, E. Amer, and T. Abuhmed. 2022. Two-stage deep learning model for alzheimer's disease detection and prediction of the mild cognitive impairment time. *Neural Computing & Applications* 34 (17):14487–509. doi: 10.1007/s00521-022-07263-9.

Eng, S. W. M., F. A. Aeschlimann, M. van Veenendaal, R. A. Berard, A. M. Rosenberg, Q. Morris, and R. S. M. Yeung. 2019. Patterns of joint involvement in juvenile idiopathic arthritis and prediction of disease course: A prospective study with multilayer non-negative matrix factorization. *PLOS Medicine* 16 (2):e1002750. doi: 10.1371/journal.pmed.1002750.

Freepik. *All the assets you need, in one place*. https://www.freepik.com/.

Goetz, C. G., B. C. Tilley, S. R. Shaftman, G. T. Stebbins, S. Fahn, P. Martinez- Martin, W. Poewe, et al. 2008. Movement disorder society-sponsored revision of the unified Parkinson's disease rating scale (MDS-UPDRS): Scale presentation and clinimetric testing results. *Movement Disorders* 23 (15):2129–70. doi: 10.1002/mds.22340.

Gunning, D., and D. Aha. 2019. DARPA's explainable artificial intelligence (XAI) program. *AI Magazine* 40 (2):44–58. doi: 10.1609/aimag.v40i2.2850.

He, K., S. Huang, and X. N. Qian. 2019. Early detection and risk assessment for chronic disease with irregular longitudinal data analysis. *Journal of Biomedical Informatics* 96:103231. doi: 10.1016/j.jbi.2019.103231.

Ho, N. H., H. J. Yang, J. Kim, D. P. Dao, H. R. Park, and S. Pant. 2022. Predicting progression of Alzheimer's disease using forward-to-backward bi-directional network with integrative imputation."*Neural networks. Neural Networks* 150:422–39. doi: 10.1016/j.neunet.2022.03.016.

Hunter, D. J., and C. Holmes. 2023. "Where medical statistics meets artificial intelligence" [in en]. *New England Journal of Medicine* 389 (13):1211–19. doi: 10.1056/NEJMra2212850.

Ito, K., S. Ahadieh, B. Corrigan, J. French, T. Fullerton, and T. Tensfeldt. 2010. Disease progression meta-analysis model in Alzheimer's disease. *Alzheimer's & Dementia* 6 (1):39–53. ISSN: 1552-5260. doi: 10.1016/j.jalz.2009.05.665.

Jiang, P., X. Wang, Q. Li, L. Jin, and S. Li. 2019. Correlation-aware sparse and low-rank constrained multi-task learning for longitudinal analysis of Alzheimer's disease. *IEEE Journal of Biomedical and Health Informatics* 23 (4):1450–56. doi: 10.1109/JBHI.2018.2885331.

Jordan, M. I., and T. M. Mitchell. 2015. Machine learning: Trends, perspectives, and prospects. *Science* 349 (6245):255–60. doi: 10.1126/science.aaa8415.

Jung, W., E. Jun, H. I. Suk, and Initiative Alzheimer's Disease Neuroimaging. 2021. Deep recurrent model for individualized prediction of Alzheimer's disease progression. *Neuroimage* 237:118143.

Kingma, D. P., and J. Ba. 2015. Adam: A method for stochastic optimization. In *3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7-9, 2015, Conference Track Proceedings*, San Diego, CA, USA, ed. Y. Bengio and Y. LeCun. doi: 10.48550/arXiv.1412.6980.

Larie, D., G. An, and R. C. Cockrell. 2021. The use of artificial neural networks to forecast the behavior of agent-based models of pathophysiology: An example utilizing an agent-based model of sepsis. *Frontiers in Physiology* 12:12. doi: 10.3389/fphys.2021.716434.

Levey, A. S., J. P. Bosch, J. B. Lewis, T. Greene, N. Rogers, and D. Roth. 1999. A more accurate method to estimate glomerular filtration rate from serum creatinine: A new prediction equation. Modification of diet in renal disease study group [in en]. *Annals of Internal Medicine* 130 (6):461–70. doi: 10.7326/0003-4819-130-6-199903160-00002.

Levey, A. S., L. A. Stevens, C. H. Schmid, Y. Lucy Zhang, A. F. Castro 3rd, H. I. Feldman, J. W. Kusek, P. Eggers, F. Van Lente, T. Greene, et al. 2009. A new equation to estimate glomerular filtration rate [in en]. *Annals of Internal Medicine* 150 (9):604–12. doi: 10.7326/0003-4819-150-9-200905050-00006.

Liu, X., J. Wang, F. Ren, and J. Kong. 2020. Group guided fused Laplacian sparse group lasso for modeling Alzheimer's disease progression. *Computational & Mathematical Methods in Medicine* 2020:1–23. doi: 10.1155/2020/4036560.

Ma, L. Y., Y. Tian, C. R. Pan, Z. L. Chen, Y. Ling, K. Ren, J. S. Li, and T. Feng. 2021. Motor progression in early-stage Parkinson's disease: A clinical prediction model and the role of cerebrospinal fluid biomarkers. *Frontiers in Aging Neuroscience* 12. doi: 10.3389/fnagi.2020.627199.

Marti-Juan, G., G. Sanroma-Guell, and G. Piella. 2020. A survey on machine and statistical learning for longitudinal analysis of neuroimaging data in Alzheimer's disease. *Computer Methods and Programs in Biomedicine* 189:189. doi: 10.1016/j.cmpb.2020.105348.

Molnar, C., ed. 2020. Local surrogate (LIME). In Interpretable machine learning; a guide for making black box models explainable. Lulu.com.

Mould, D. R. 2012. Models for disease progression: New approaches and uses. *Clinical Pharmacology & Therapeutics* 92 (1):125–31. doi: 10.1038/clpt.2012.53.

Nauta, M., J. Trienes, S. Pathak, E. Nguyen, M. Peters, Y. Schmitt, J. Schlötterer, M. van Keulen, and C. Seifert. 2023. From anecdotal evidence to quantitative evaluation methods: A systematic review on evaluating explainable AI. *ACM Computing Surveys* 55 (13s):1–42. doi: 10.1145/3583558.

Nazha, A., R. Komrokji, M. Meggendorfer, X. Jia, N. Radakovich, J. Shreve, C. Beau Hilton, Y. Nagata, B. K. Hamilton, S. Mukherjee, et al. 2021. Personalized prediction model to risk stratify patients with myelodysplastic syndromes. *Journal of Clinical Oncology* 39 (33):3737–46. doi: 10.1200/JCO.20.02810.

Papaiz, F., Jr., M. E. T. Dourado, R. A. D. M. Valentim, A. H. F. de Morais, and J. P. Arrais. 2022. Machine learning solutions applied to amyotrophic lateral sclerosis prognosis: A review. *Frontiers in Computer Science* 4. doi: 10.3389/fcomp.2022.869140.

Post, T. M., J. I. Freijer, J. DeJongh, and M. Danhof. 2005. Disease system analysis: Basic disease progression models in degenerative disease. *Pharmaceutical Research* 22 (7):1038–49. doi: 10.1007/s11095-005-5641-5.

Pukelsheim, F. 1994. The three sigma rule. *American Statistician* 48 (2):88–91. doi: 10.1080/00031305.1994.10476030.

Rad, B., F. Song, V. Jacob, and Y. Diao. 2021. Explainable anomaly detection on high-dimensional time series data. In *Proceedings of the 15th ACM International Conference on Distributed and Event-Based Systems*, *2–14. DEBS '21. Virtual Event*, Italy: Association for Computing Machinery. ISBN: 9781450385558. doi: 10.1145/3465480.3468292.

Ribeiro, M. T., S. Singh, C. Guestrin, and ISBN: 9781450342322. https://doi.org/10.1145/2939672.2939778.2016. "Why should I trust you?": Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, *1135–1144. KDD '16*, San Francisco, CA, USA: Association for Computing Machinery.

Sadiq, M. U., K. Kwak, E. Dayan, and Initiative for the Alzheimer's Disease Neuroimaging. 2022. Model-based stratification of progression along the Alzheimer disease continuum highlights the centrality of biomarker synergies. *Alzheimer's Research & Therapy* 14 (1). doi: 10.1186/s13195-021-00941-1.

Schmidt-Erfurth, U., S. M. Waldstein, S. Klimscha, A. Sadeghipour, X. Hu, B. S. Gerendas, A. Osborne, and H. Bogunovic´. 2018. Prediction of individual disease conversion in early AMD using artificial intelligence. *Inves- Tigative Ophthalmology and Visual Science* 59 (8):3199–208. doi: 10.1167/iovs.18-24106.

Schulam, P., and S. Saria. 2016. Integrative analysis using coupled latent variable models for individualizing prognoses. *Journal of Machine Learning Research* 17 (1 (January):8244–78. ISSN: 1532-4435.

Seccia, R., S. Romano, M. Salvetti, A. Crisanti, L. Palagi, and F. Grassi. 2021. Machine learning use for prognostic purposes in multiple sclerosis. *Life* 11 (2):1–18. doi: 10.3390/life11020122.

Sharma, R., H. Anand, Y. Badr, and R. G. Qiu. 2021. Time-to-event prediction using survival analysis methods for Alzheimer's disease progression. *Alzheimer's & Dementia: Translational Research & Clinical Interventions* 7 (1). doi: 10.1002/trc2.12229.

Su, W., X. Wang, and R. D. Szczesniak. 2020. Flexible link functions in a joint hierarchical Gaussian process model. *Biometrics Bulletin* 77 (2):754–64. doi: 10.1111/biom.13291.

Sun, C., S. Hong, M. Song, H. Li, and Z. Wang. 2021. Predicting COVID-19 disease progression and patient outcomes based on temporal deep learning. *BMC Medical Informatics & Decision Making* 21 (1). doi:10.1186/s12911-020-01359-9.

Sun, C., H. Li, R. E. Mills, and Y. Guan. 2019. Prognostic model for multiple myeloma progression integrating gene expression and clinical features. *GigaScience* 8 (12). doi: 10.1093/gigascience/giz153.

Taghavi, M., F. Staal, F. G. Munoz, F. Imani, D. B. Meek, R. Simões, L. G. Klompenhouwer, U. A. van der Heide, R. G. H. Beets-Tan, and M. Maas. 2021. CT-Based radiomics analysis

before thermal ablation to predict local tumor progression for colorectal liver metastases. *Cardiovascular and Interventional Radiology* 44 (6):913–20. doi: 10.1007/s00270-020-02735-8.

Tjoa, E., and C. Guan. 2021. A survey on explainable artificial intelligence (XAI): Toward medical XAI. *IEEE Transactions on Neural Networks and Learning Systems* 32 (11):4793–813. doi: 10.1109/TNNLS.2020.3027314.

Van Den Eeden, S. K., C. M. Tanner, A. L. Bernstein, R. D. Fross, A. Leimpeter, D. A. Bloch, and L. M. Nelson. 2003. Incidence of Parkinson's disease: Variation by age, gender, and race/ethnicity. *American Journal of Epidemiology* 157 (11 (June)):1015–22. ISSN: 0002-9262. doi: 10.1093/aje/kwg068.

Vayena, E., A. Blasimme, and I. Glenn Cohen. 2018. Machine learning in medicine: Addressing ethical challenges. *PLOS Medicine* 15 (11):1–4. doi: 10.1371/journal.pmed.1002689.

Wen, C. P., C. H. Chang, M. K. Tsai, J. H. Lee, P. J. Lu, S. P. Tsai, C. Wen, C.H. Chen, C.W. Kao, C.K. Tsao, et al. 2017. Diabetes with early kidney involvement may shorten life expectancy by 16 years. *Cited By: 84; All Open Access, Bronze Open Access, Kidney International* 92 (2):388–96. doi: 10.1016/j.kint.2017.01.030.

Yılmaz Acar, Z., F. Başçiftçi, and A. H. Ekmekci. 2022. Future activity prediction of multiple sclerosis with 3D MRI using 3D discrete wavelet transform. *Biomedical Signal Processing and Control* 78:103940. doi: 10.1016/j.bspc.2022.103940.

Zhang, J., J. Ning, X. Huang, and R. Li. 2021. On the time-varying predictive performance of longitudinal biomarkers: Measure and estimation. *Statistics in Medicine* 40 (23):5065–77. doi: 10.1002/sim.9111.

Zheng, Y., and X. Hu. 2020. Healthcare predictive analytics for disease progression: A longitudinal data fusion approach. *Journal of Intelligent Information Systems* 55 (2):351–69. doi:10.1007/s10844-020-00606-9.

Appendix

A Imputation method comparison results

Table A1. The predictive performance results for the PPMI dataset, reported in MSE. Bold indicates the best score for the imputation type across all folds and seeds, italic indicates the worst score.

Seed	Imputation type	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average	Std. dev.
787	Population mean	187.13	188.88	212.74	221.23	235.60	209.12	18.73
787	Event mean	199.37	216.98	217.98	290.36	235.04	231.95	31.31
787	Forward/backward filling	180.94	213.84	232.54	*308.84*	290.86	245.40	47.77
1998	Population mean	173.82	263.44	209.38	*448.41*	212.91	261.59	97.68
1998	Event mean	179.43	347.52	215.52	165.40	172.64	216.10	67.93
1998	Forward/backward filling	222.31	253.87	250.84	237.19	194.40	231.72	21.76
959143	Population mean	185.23	197.35	**139.20**	163.05	212.67	179.50	25.87
959143	Event mean	169.94	164.26	**137.44**	170.36	252.94	178.99	38.91
959143	Forward/backward filling	174.85	157.14	143.11	**130.08**	181.54	157.34	19.17
25	Population mean	274.09	265.23	304.78	372.65	301.4	303.63	37.73
25	Event mean	312.20	289.64	330.55	325.54	*370.75*	325.74	26.59
25	Forward/backward filling	255.71	249.09	237.76	287.00	265.50	259.01	16.65

Table A2. The predictive performance results for the diabetes dataset, reported in MSE. Bold indicates the best score for the imputation type across all folds and seeds, italic indicates the worst score.

Seed	Imputation type	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average	Std. dev.
787	Population mean	42.71	**41.34**	44.12	42.55	41.98	42.54	0.93
787	Event mean	42.81	**41.15**	44.35	42.07	42.48	42.57	1.05
787	Forward/backward filling	42.52	41.83	44.41	**41.63**	42.19	42.52	0.99
1998	Population mean	46.56	45.74	46.35	45.02	45.41	45.82	0.57
1998	Event mean	46.57	45.13	45.97	45.10	44.83	45.52	0.65
1998	Forward/backward filling	45.98	44.60	44.80	43.70	44.65	44.75	0.73
959143	Population mean	41.89	42.24	42.17	42.00	45.48	42.76	1.37
959143	Event mean	41.59	42.40	41.49	42.02	44.96	42.49	1.28
959143	Forward/backward filling	42.39	43.23	42.38	42.64	41.77	42.48	0.47
25	Population mean	45.07	45.14	45.42	*47.08*	43.94	45.33	1.01
25	Event mean	46.12	45.28	45.56	*47.55*	44.58	45.82	1.00
25	Forward/backward filling	*47.09*	45.05	46.74	47.08	44.11	46.01	1.21

B PPMI case study feature correlations

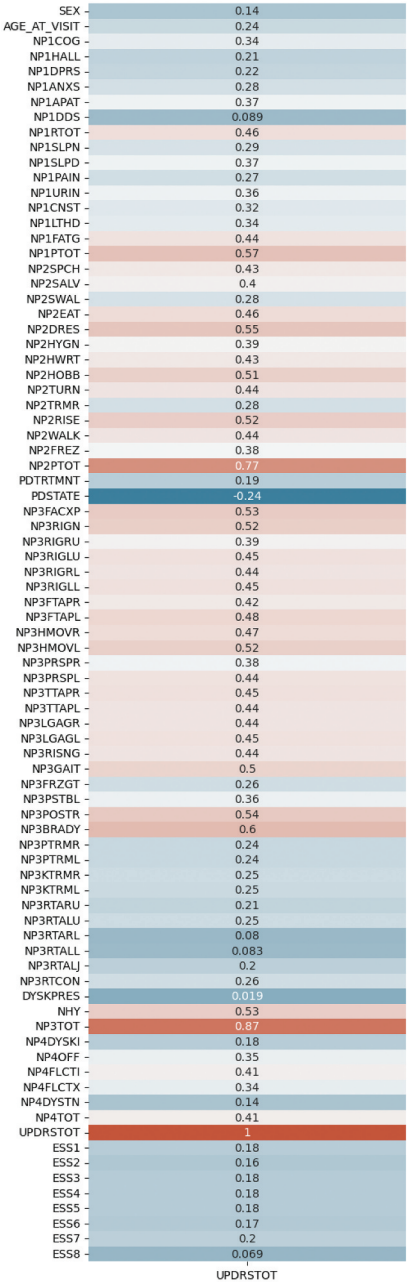


Figure B1. A plot of the correlation between the PPMI features and the target UPDRS total score.